WEEK 1

SOFTWARE DEVELOPMENT TOOLS AND ENVIRONMENTS

UNLIMITED

Presented by **Asst. Prof. Dr. Tuchsanai Ploysuwan**

- Git was developed in 2005 by **Linus Torvalds**
- Git is **Version control system** is a system that records changes to a file or set file over time so that you. can restore specific version later
- Git is a **Distributed Version Control System**

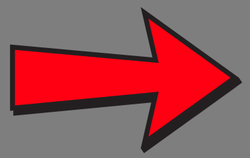
Git – What and Why

Before Version Control System

Before Version Control System

Before Version Control System

Version 1.0



running code version 1.0

Before Version Control System

Version 1.0

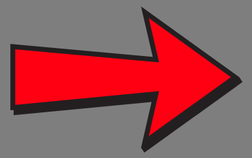
Version 2.0

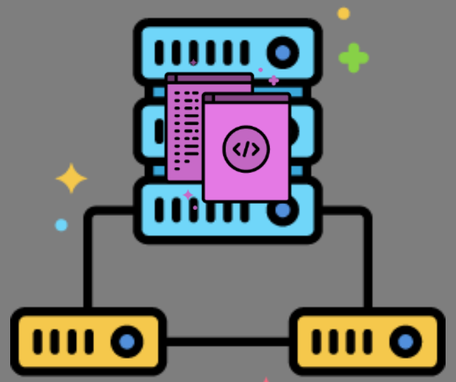
running code version 1.0

Before Version Control System

Version 1.0

Version 2.0





running code version 2.0

Before Version Control System

Version 1.0

Version 2.0

running code version 2.0

Before Version Control System

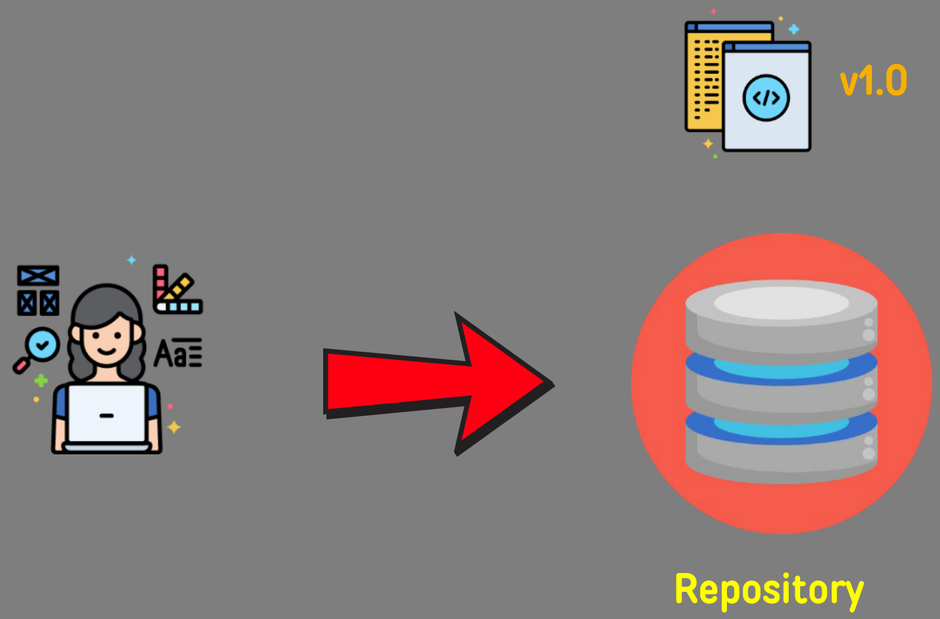
Version 1.0

Version 2.0

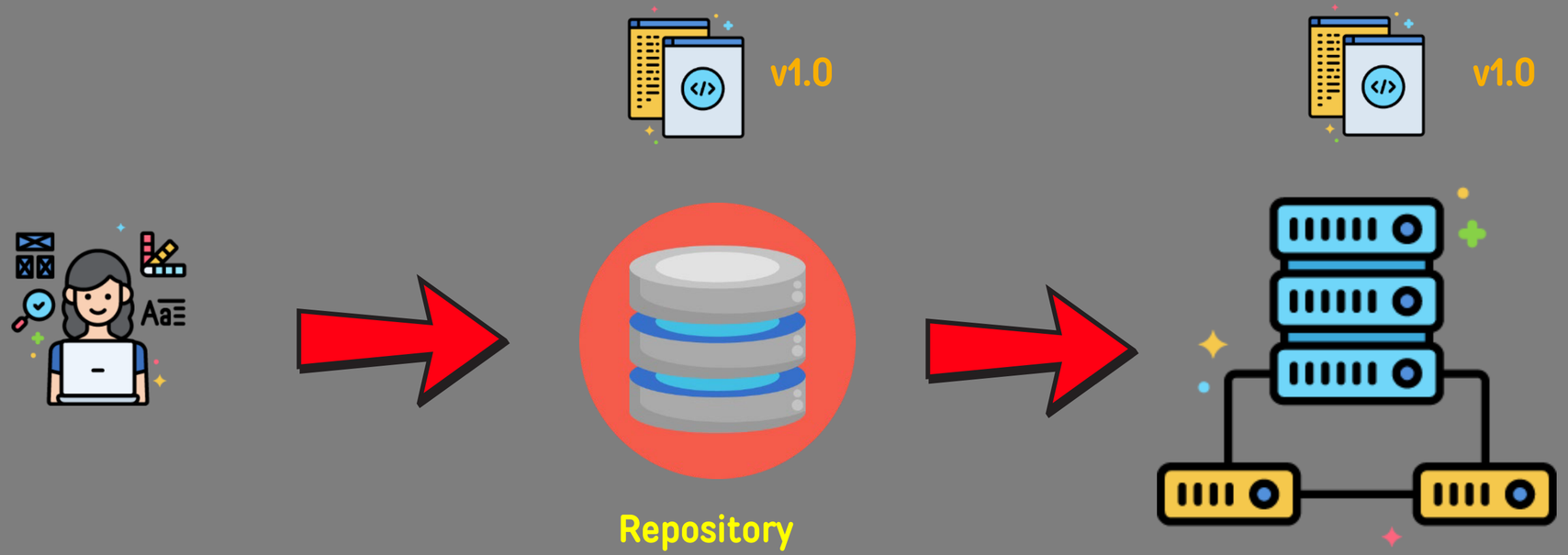
running code version 2.0

- Rollback is time consuming
- No audit tracking
- Not scalable for large teams

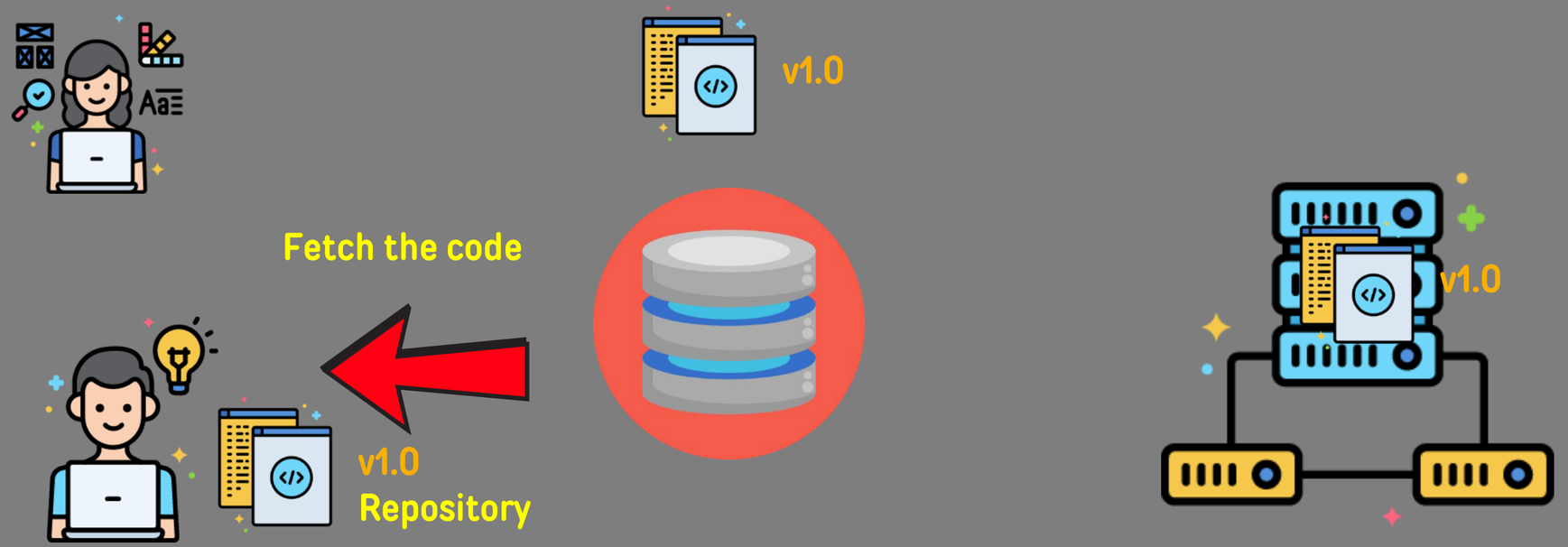
Version Control System



Version Control System



Version Control System

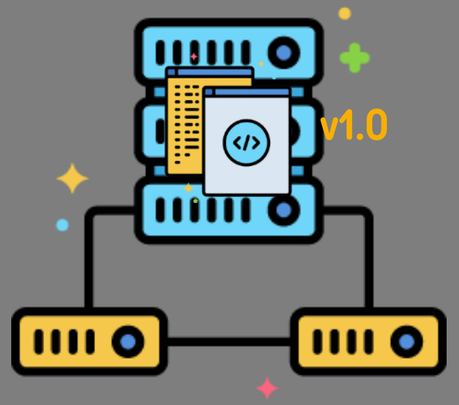


Version Control System

v1.0

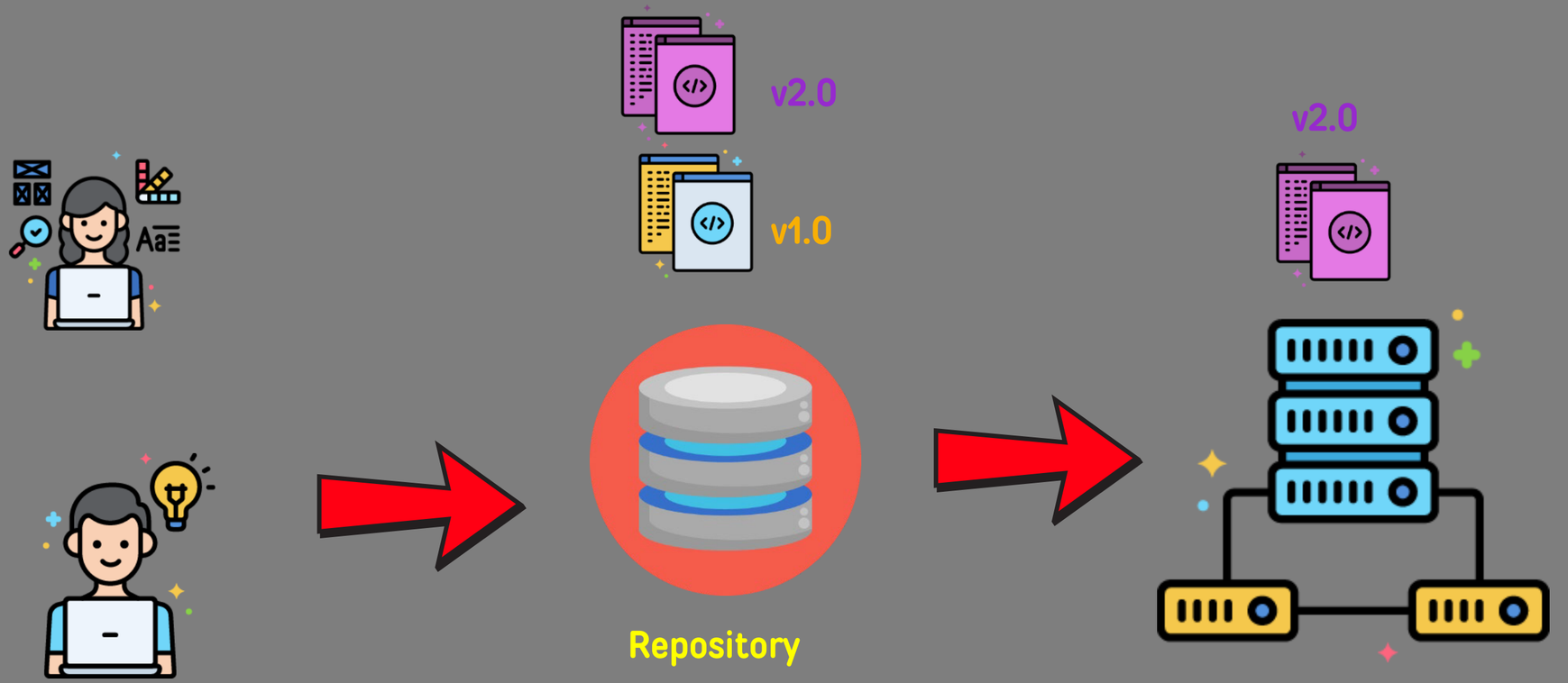
v2.0

Repository



v1.0

Version Control System



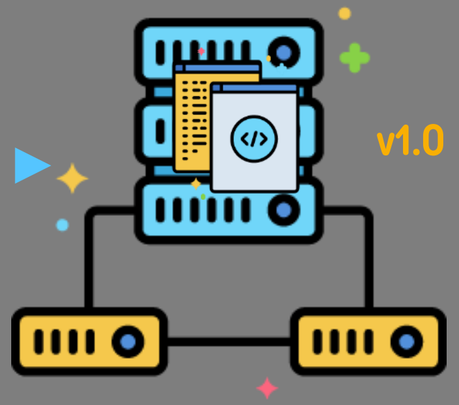
Version Control System

Repository

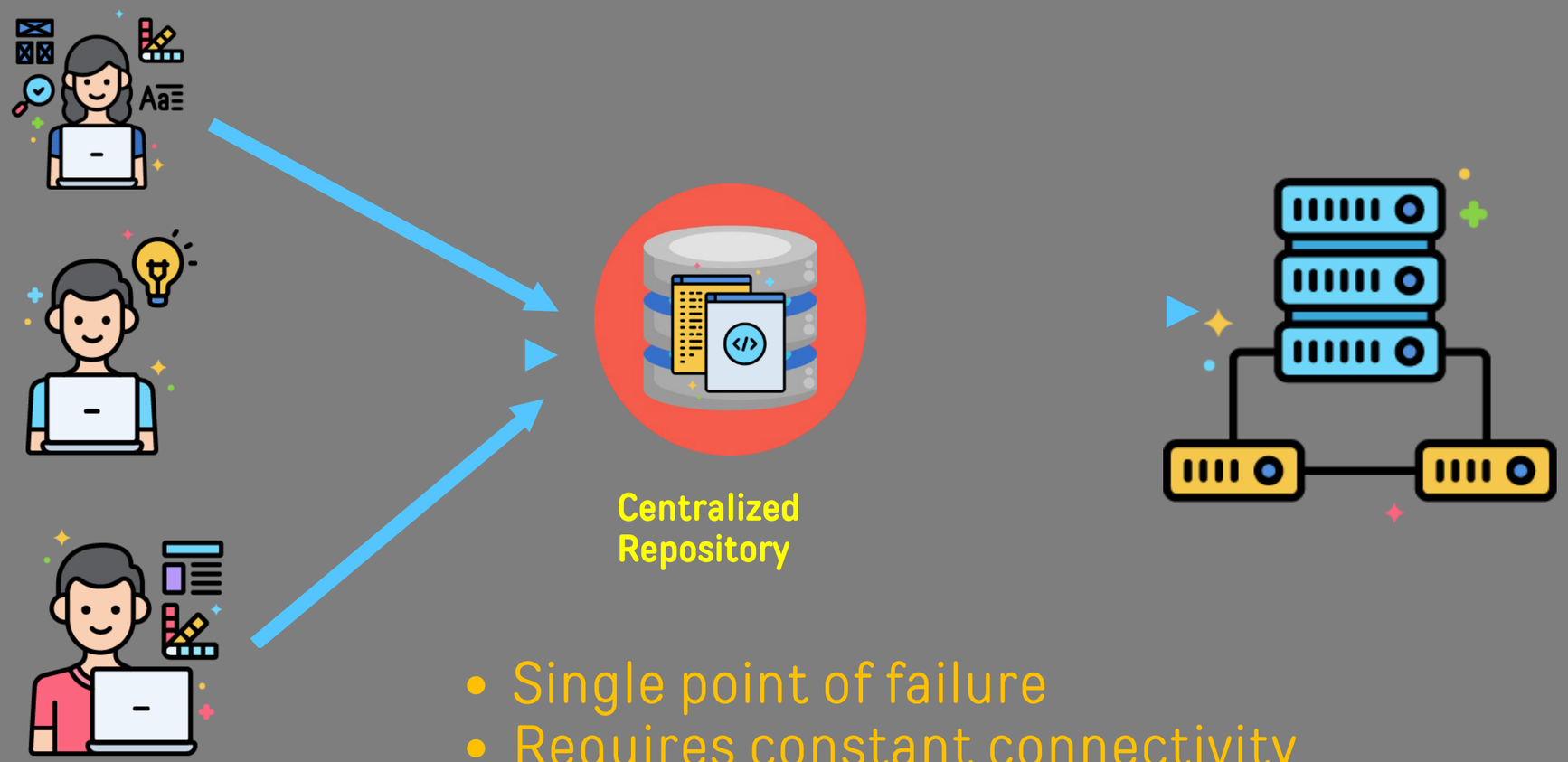
Version Control System

Why Git?

- Distributed

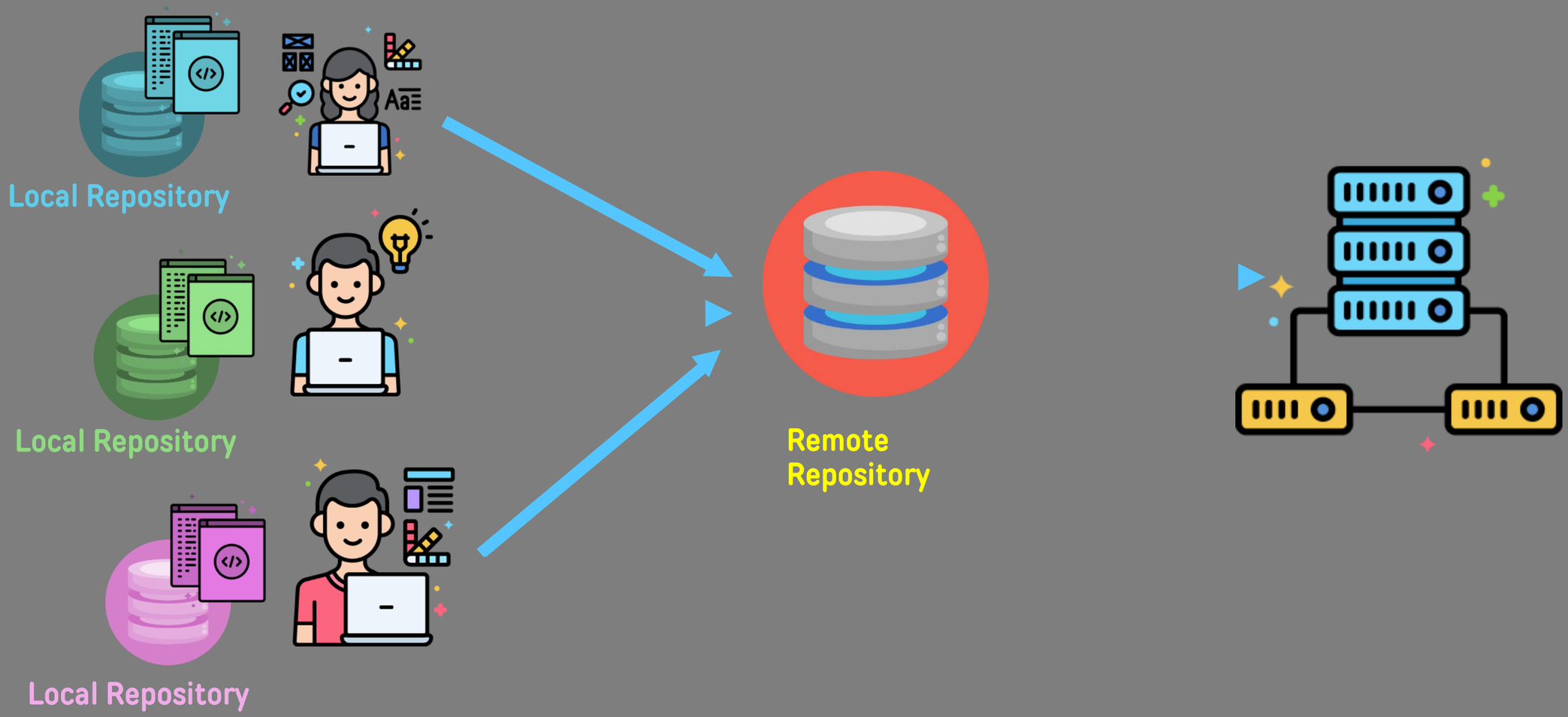


Centralized Version Control System



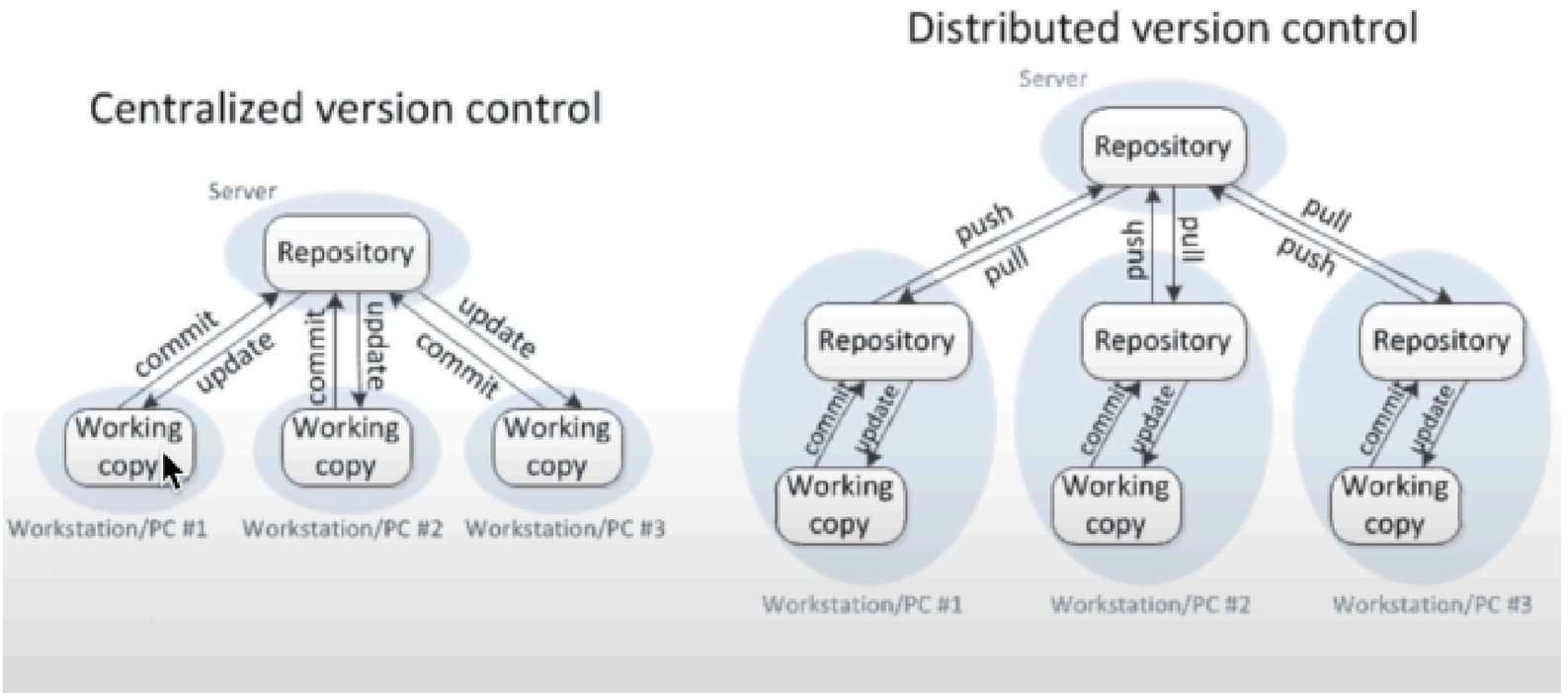
- Single point of failure
- Requires constant connectivity
- E.G – Subversion, Endevor

Distributed Version Control System

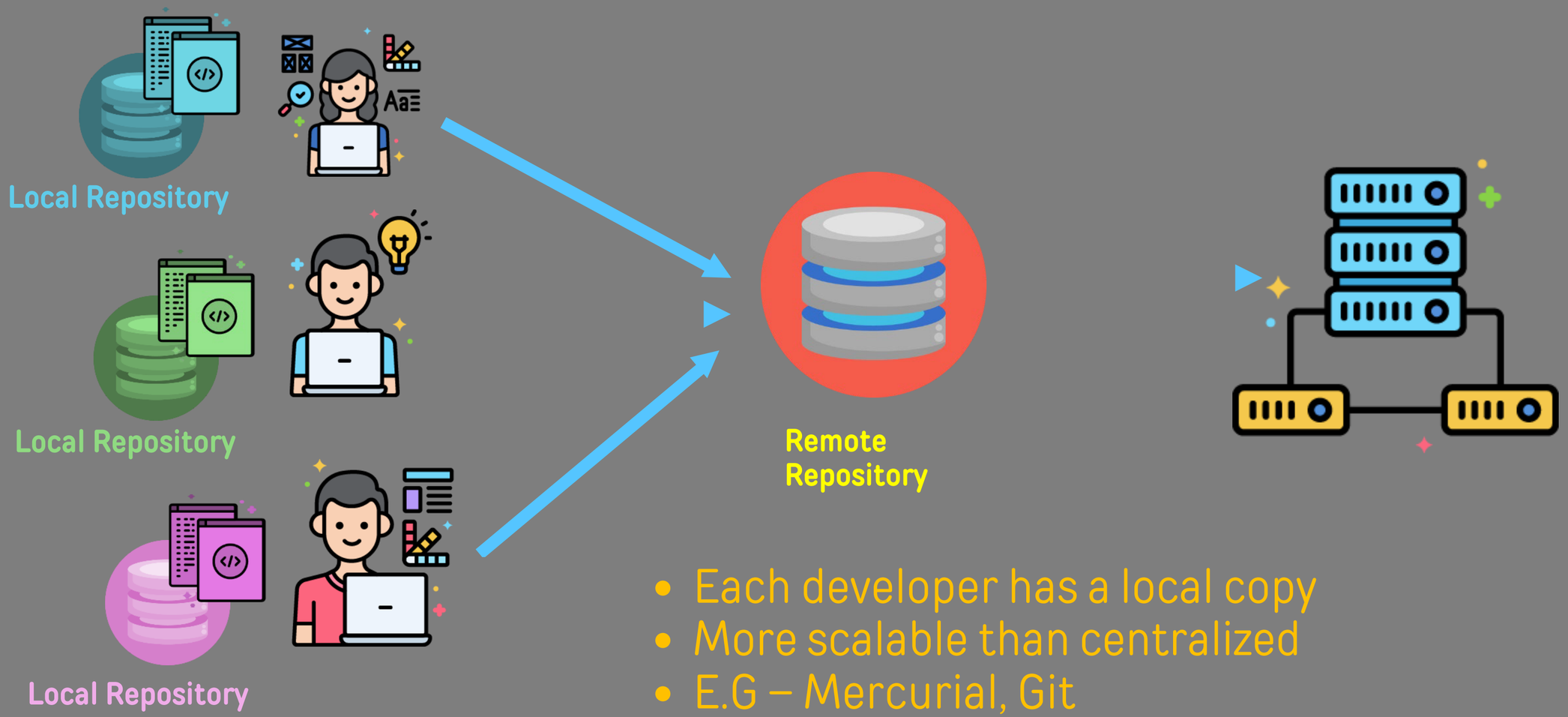


Centralized vs Distributed

Version Control



Distributed Version Control System



- Each developer has a local copy
- More scalable than centralized
- E.G – Mercurial, Git

Version Control System

Why Git?

- Distributed
- Performant
- Detailed audit tracking
- Open source
  - Free!
  - Implemented with Kubernetes GitOps, integration with Jenkins and other DevOps tools
  - GitHub, GitLab, Code Commit are all based on Git

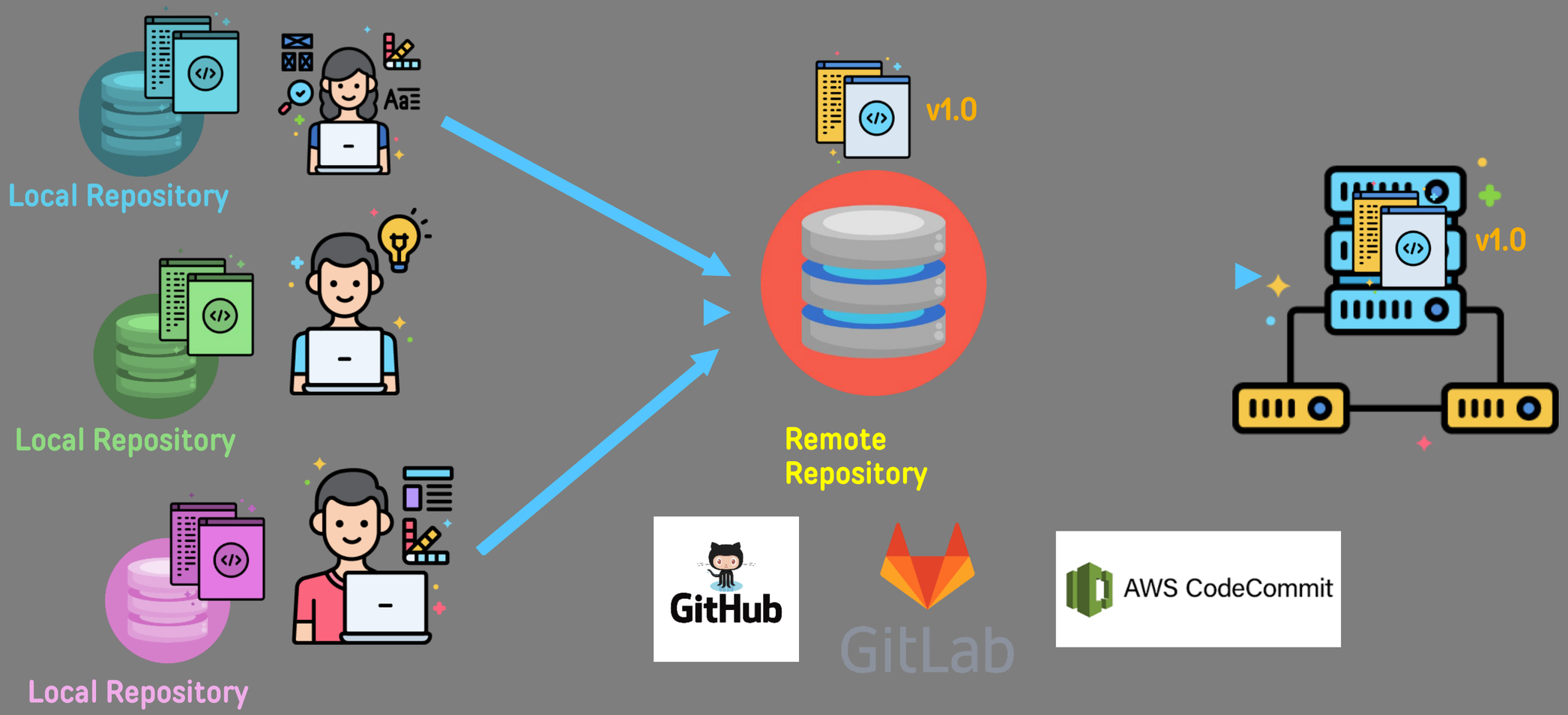
Git vs GitHub

Git Vs. GitHub

- Version Control System
- Installed locally on the system
- Created in 2005, by Linus Torvalds
- Open source, and used in multiple cloud repository services

- Git repository hosting services with other features
- Runs on the cloud
- Created in 2008, currently owned by Microsoft
- Not open source, have free and paid tiers

Distributed Version Control System



Configure Git

การตั้งค่า Git (Configure)

ทำไมต้องตั้งค่าก่อนใช้งาน?

ทุกครั้งที่เรา commit — Git จะ **แนบชื่อ (user.name) และอีเมล (user.email) ของผู้เขียนติดไปกับ commit นั้นถาวร** เพื่อบอกว่า "ใครเป็นคนแก้โค้ดชุดนี้" ถ้ายังไม่ตั้งค่า Git จะไม่ยอมให้ commit และแจ้ง error ว่า `Please tell me who you are`

🪪

ระบุตัวตนผู้เขียน

ชื่อ + อีเมลถูกฝังในทุก commit ทำให้ตามได้ว่าใครแก้อะไร เมื่อไหร่ (ดูได้จาก `git log`)

🔗

เชื่อมกับ GitHub

ถ้าอีเมลตรงกับที่ผูกไว้ใน GitHub account — commit จะโชว์รูปโปรไฟล์และนับเป็น contribution (ช่องเขียว) ของเรา

1️⃣

ตั้งครั้งเดียวพอ

ค่าแบบ `--global` ตั้งครั้งเดียวใช้ได้กับทุก repo บนเครื่อง ไม่ต้องตั้งใหม่ทุกโปรเจกต์

ขั้นที่ 1 — ตั้งชื่อและอีเมล (บังคับ)

```
# ตั้งค่าชื่อและอีเมล (--global = ใช้กับทุก repo บนเครื่อง)
git config --global user.name "Your Name"
git config --global user.email "you@example.com"

# ตัวอย่างจริง (สมมติ)
git config --global user.name "somchai-dev"
git config --global user.email "somchai.dev@example.com"
```

ขั้นที่ 2 — ตั้งค่าเสริมที่แนะนำ (Optional แต่ควรทำ)

```
# ให้ branch แรกชื่อ main (แทน master) ตรงกับมาตรฐาน GitHub
git config --global init.defaultBranch main

# ตั้ง VS Code เป็น editor เวลา Git ต้องให้เราพิมพ์ข้อความ (เช่น commit message)
git config --global core.editor "code --wait"

# เปิดสีในผลลัพธ์คำสั่ง อ่าน git status / git diff ง่ายขึ้น
git config --global color.ui auto

# จำรหัสผ่าน/token ไว้ ไม่ต้องกรอกซ้ำทุกครั้งที่ push/pull
git config --global credential.helper store
```

ขั้นที่ 3 — ตรวจสอบการตั้งค่า

```
# ดูค่า config ทั้งหมด (กด q เพื่อออก)
git config --list              # หรือย่อเป็น  git config -l
>> user.name=somchai-dev
>> user.email=somchai.dev@example.com
>> init.defaultbranch=main
>> color.ui=auto

# ดูทีละค่า – เช็กเฉพาะตัวที่สงสัย
git config user.name
>> somchai-dev
git config user.email
>> somchai.dev@example.com

# ดูว่าค่าแต่ละตัวถูกตั้งจากไฟล์ไหน (มีประโยชน์มากเวลาค่า local ทับ global)
git config --list --show-origin
>> file:/home/somchai/.gitconfig      user.name=somchai-dev
>> file:.git/config                   user.name=work-account
```

สลับไปใช้ GitHub อีก account

กรณีมีหลาย account (เช่น account ส่วนตัว กับ account ของมหาวิทยาลัย) ต้องเปลี่ยนทั้ง **ตัวตนใน commit** (user.name / user.email) และ **ตัวตนที่ใช้ login** (credential.username):

```
# เปลี่ยนตัวตนที่แนบใน commit
git config --global user.name "somchai-uni"
git config --global user.email "somchai.j@university-example.ac.th"

# เปลี่ยน username ที่ใช้ยืนยันตัวตนกับ GitHub ตอน push/pull
git config --global credential.username "somchai-uni"
```

แก้ปัญหา: commit / push ไม่ได้เพราะค้าง account เก่า

อาการที่พบบ่อยในห้อง Lab: เครื่องเคย login ด้วย account ของคนก่อนหน้า ทำให้ push แล้วขึ้น `Permission denied` หรือ 403 — วิธีแก้คือล้าง config เก่าออกก่อน แล้วตั้งใหม่:

```
# 1) ลบ config ตัวตนเก่าออกทั้งหมด (--unset = ลบค่านั้นทิ้ง)
git config --global --unset user.name
git config --global --unset user.email
git config --global --unset credential.username

# 2) ถ้า repo ชี้ remote ของคนเก่าอยู่ ให้ถอด origin ออก
git remote -v                # เช็กก่อนว่า origin ชี้ไปที่ไหน
>> origin  https://github.com/old-user/old-repo.git (fetch)
git remote remove origin     # ลบ origin เก่า แล้วค่อย git remote add ใหม่

# 3) ตั้งค่า account ของเราใหม่ตามขั้นที่ 1
```

Creating a Git Repository

Starting with Git

- How can we create a Git Repository?
  - **git init**
    - This command initializes a Git Repository on your local machine.
    - You only need to run this command once per project.
  - **git status**
    - This command will report back the status of your Git repository.

Starting with Git

- How can we create a Git Repository?
  - Upon creating a repository with **git init** you will create a hidden .git file.
  - The .git file is a hidden file that manages the versioning of the files inside the Git repository.

Starting with Git

- Git inside a Folder/Directory:
  - Upon creating a Git Repository, all the folders/directories inside the top level Git Repository will also be part of that Repository, meaning all the changes are tracked.

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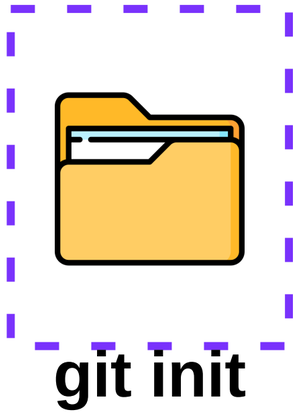

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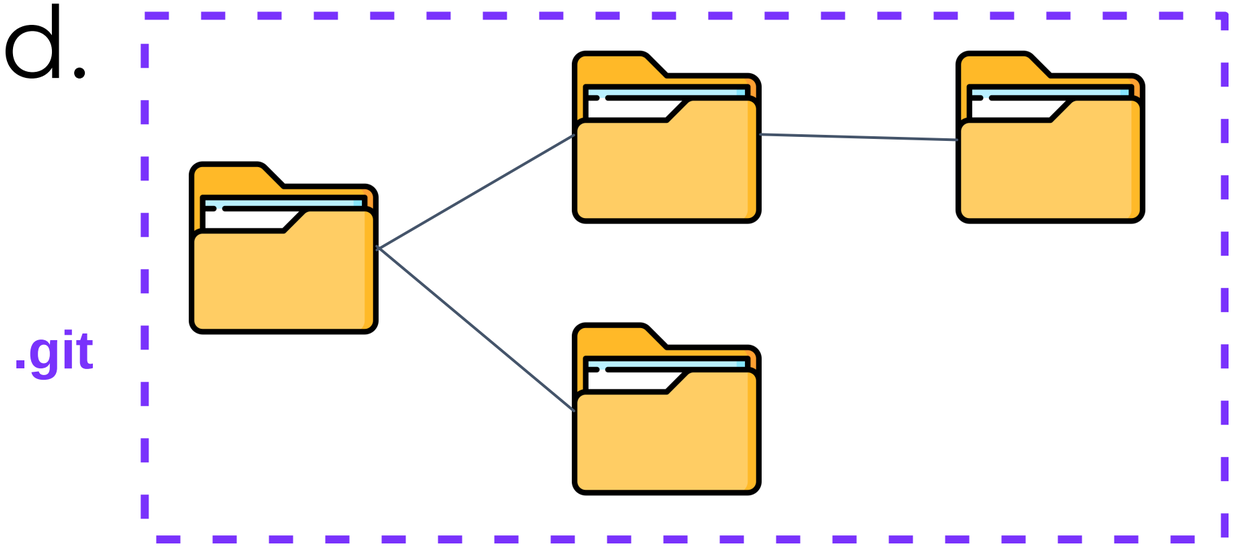

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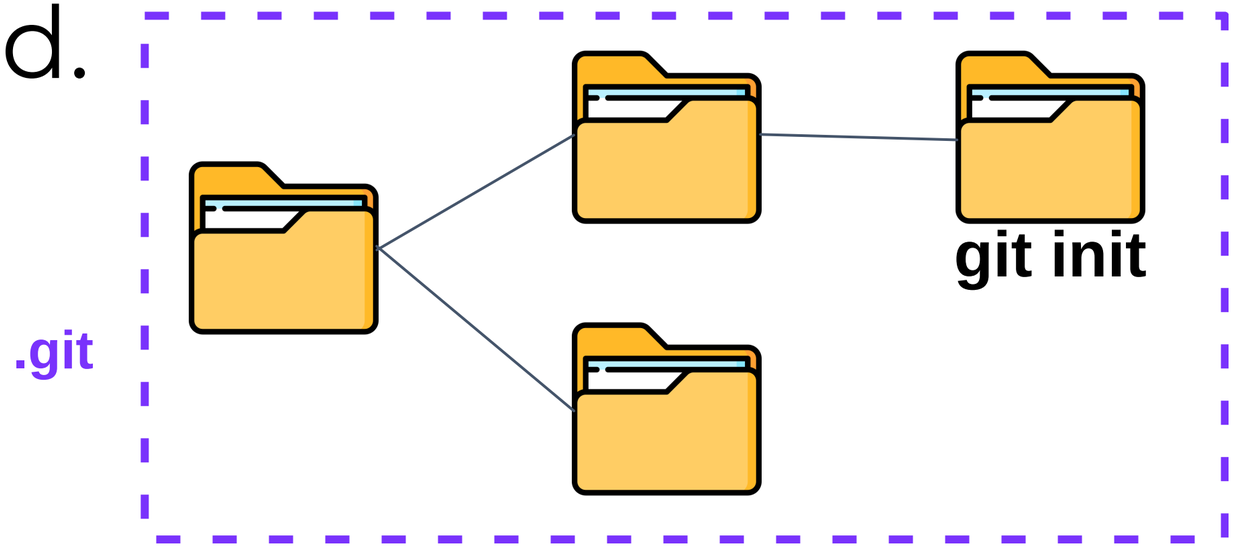

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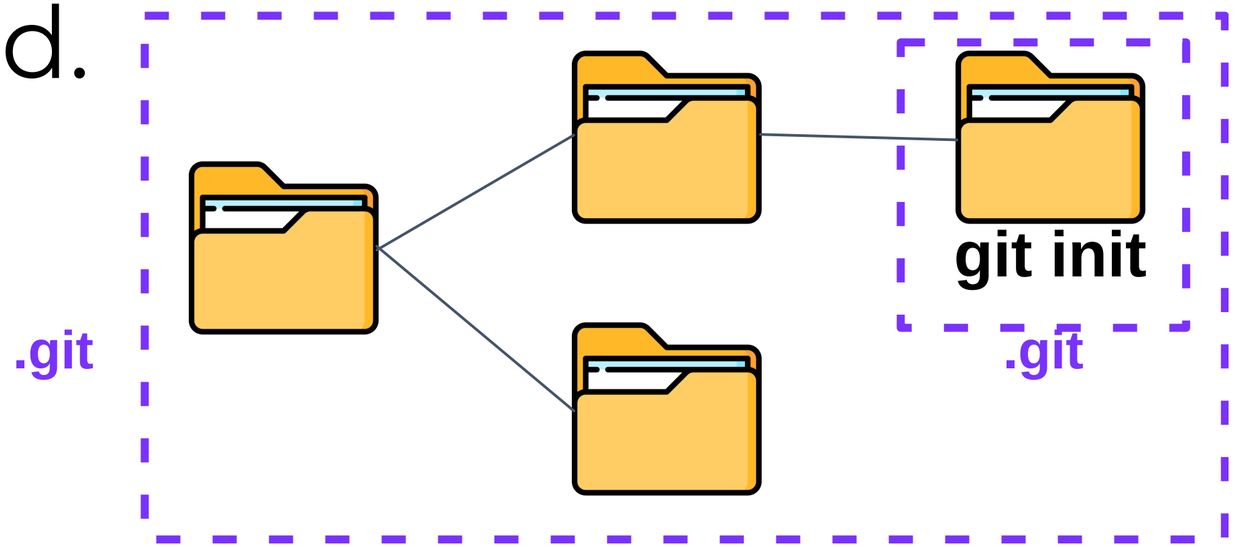

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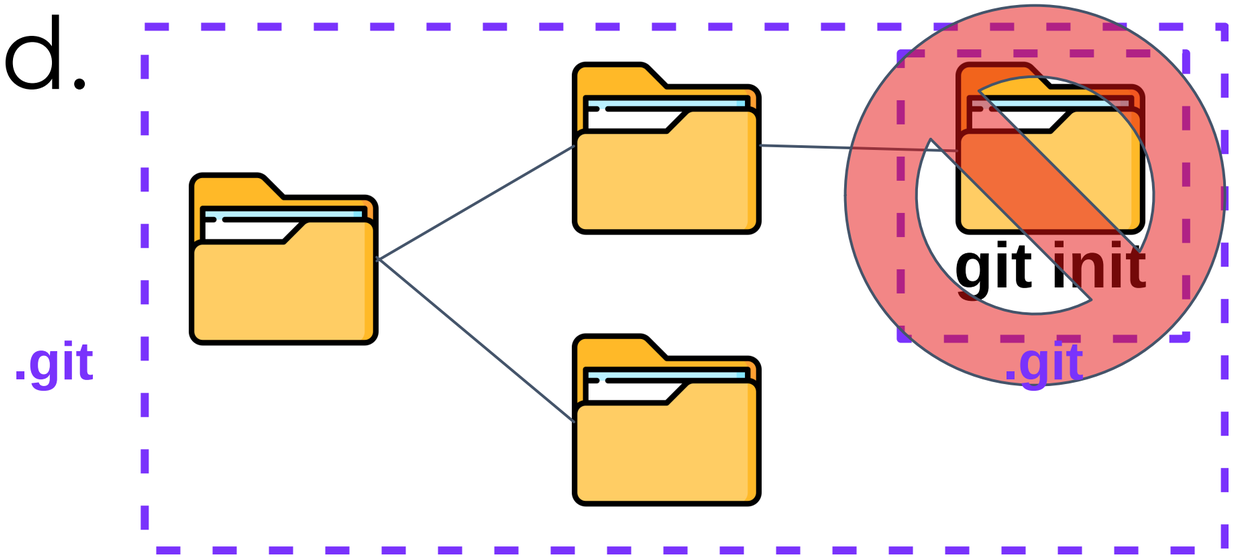

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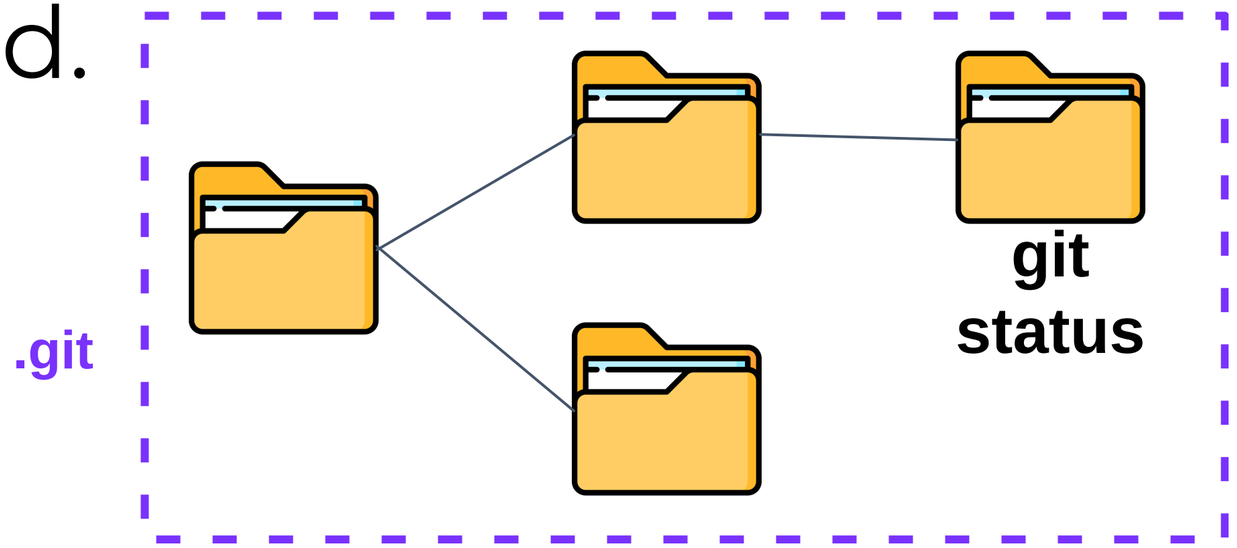
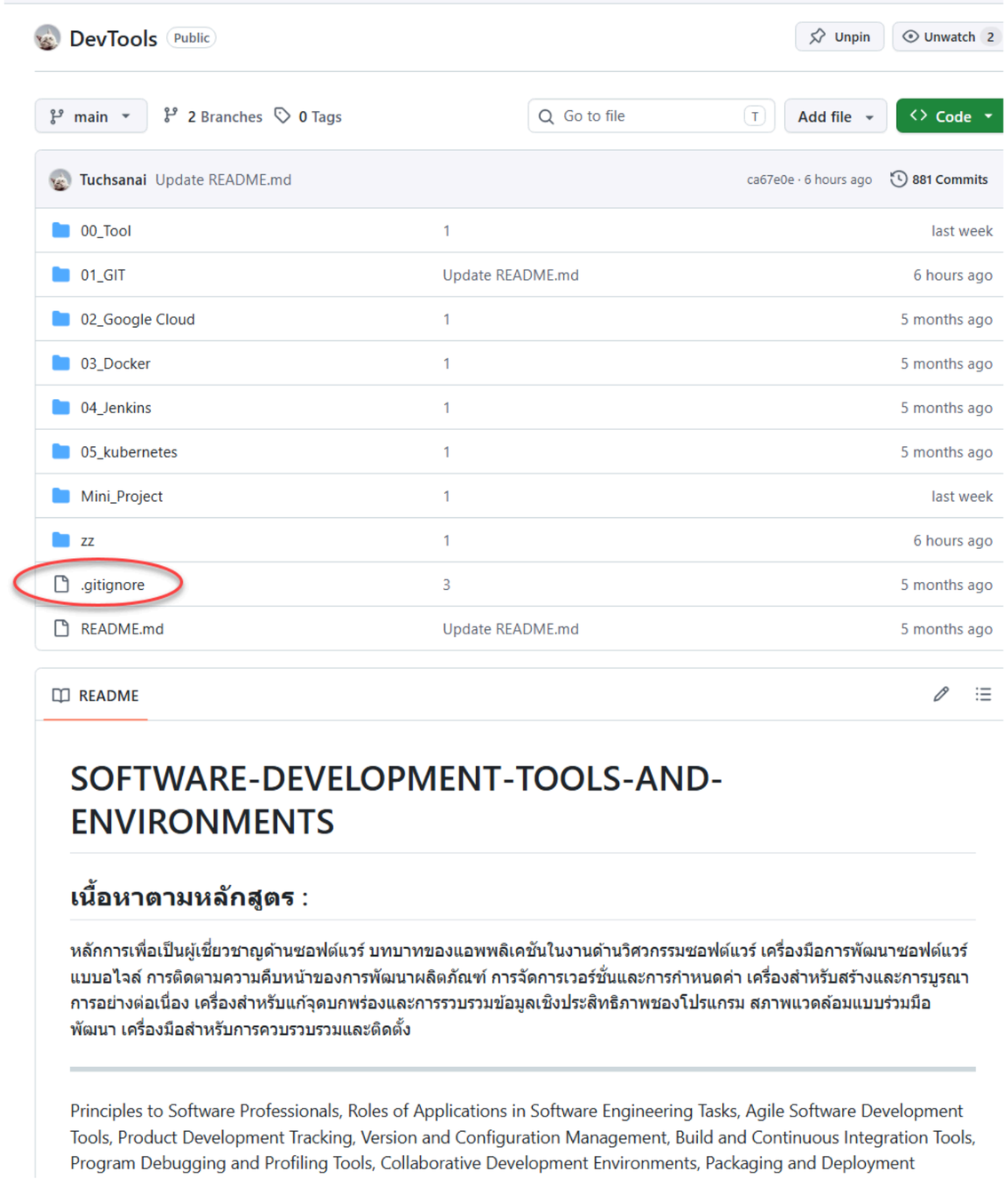


Ignoring Files

We can tell Git which files and directories to ignore in a given repository, using a .gitignore file. This is useful for files you know you NEVER want to commit, including: Secrets, API keys, credentials, etc. Operating System files (.DS_Store on Mac) Log files Dependencies & packages



main DevTools / .gitignore

Code Blame 229 lines (182 loc) · 4.2 KB

```
# IPython
profile_default/
ipython_config.py

# Remove previous ipynb_checkpoints
#   git rm -r .ipynb_checkpoints/

### macOS ###
# General
.DS_Store
*.DS_Store
.AppleDouble
.LSOverride

# Icon must end with two \r
Icon


# Thumbnails
._*

# Files that might appear in the root of a volume
.DocumentRevisions-V100
.fseventsd
.Spotlight-V100
.TemporaryItems
.Trashes
.VolumeIcon.icns
.com.apple.timemachine.donotpresent

# Directories potentially created on remote AFP share
.AppleDB
.AppleDesktop
Network Trash Folder
Temporary Items
.apdisk

### macOS Patch ###
# iCloud generated files
*.icloud

### Python ###
# Byte-compiled / optimized / DLL files
__pycache__/
*.py[cod]
*$py.class

# C extensions
*.so

# Distribution / packaging
.Python
build/
develop-eggs/
```

สรุปกฎการเขียน `.gitignore`

หลักการพื้นฐาน

- 1 บรรทัด = 1 pattern
- บรรทัดว่างไม่มีผล ใช้แบ่งกลุ่มให้อ่านง่าย
- `#` ขึ้นต้นบรรทัด = comment
- ถ้าไฟล์ถูก track ไปแล้ว การเพิ่มใน `.gitignore` จะไม่มีผล ต้อง `git rm --cached <file>` ก่อน

Pattern หลัก ๆ

Pattern	ความหมาย
`*.log`	ไฟล์ .log ทุกไฟล์ ทุก directory
`build/`	ลงท้ายด้วย `/` = match เฉพาะ directory ชื่อ build (ทุกระดับ)
`/config.py`	ขึ้นต้นด้วย `/` = เฉพาะ root ของ repo เท่านั้น
`debug?.txt`	`?` = ตัวอักษร 1 ตัวใด ๆ (debug1.txt, debugA.txt)
`[0-9].txt`	character range
`!important.log`	`!` = ยกเว้น (negate) ไม่ ignore ไฟล์นี้

กฎของ `**` (double asterisk)

- `**/logs` — match `logs` ที่ระดับไหนก็ได้
- `logs/**` — ทุกอย่างภายใน `logs/`
- `a/**/b` — match `a/b`, `a/x/b`, `a/x/y/b` (ข้ามกี่ชั้นก็ได้)

กฎสำคัญเรื่อง negation (`!`)

- สำดับบรรทัดมีผล — pattern ที่มาทีหลังชนะ
- ⚠️ ถ้า directory แม่ถูก ignore ไปแล้ว จะ `!` ไฟล์ข้างในไม่ได้ ต้องเปิด directory ก่อน เช่น

.gitignore

Create a file called .gitignore in the root of a repository.Inside the file, we can write patterns to tell Git which files & folders to ignore:
.DS_Store will ignore files named .DS_Store
folderName/ will ignore an entire directory
*.log will ignore any files with the .log extension

https://www.toptal.com/developers/gitignore

Private Repositories and Tokens

Day 1 - Starting with Git

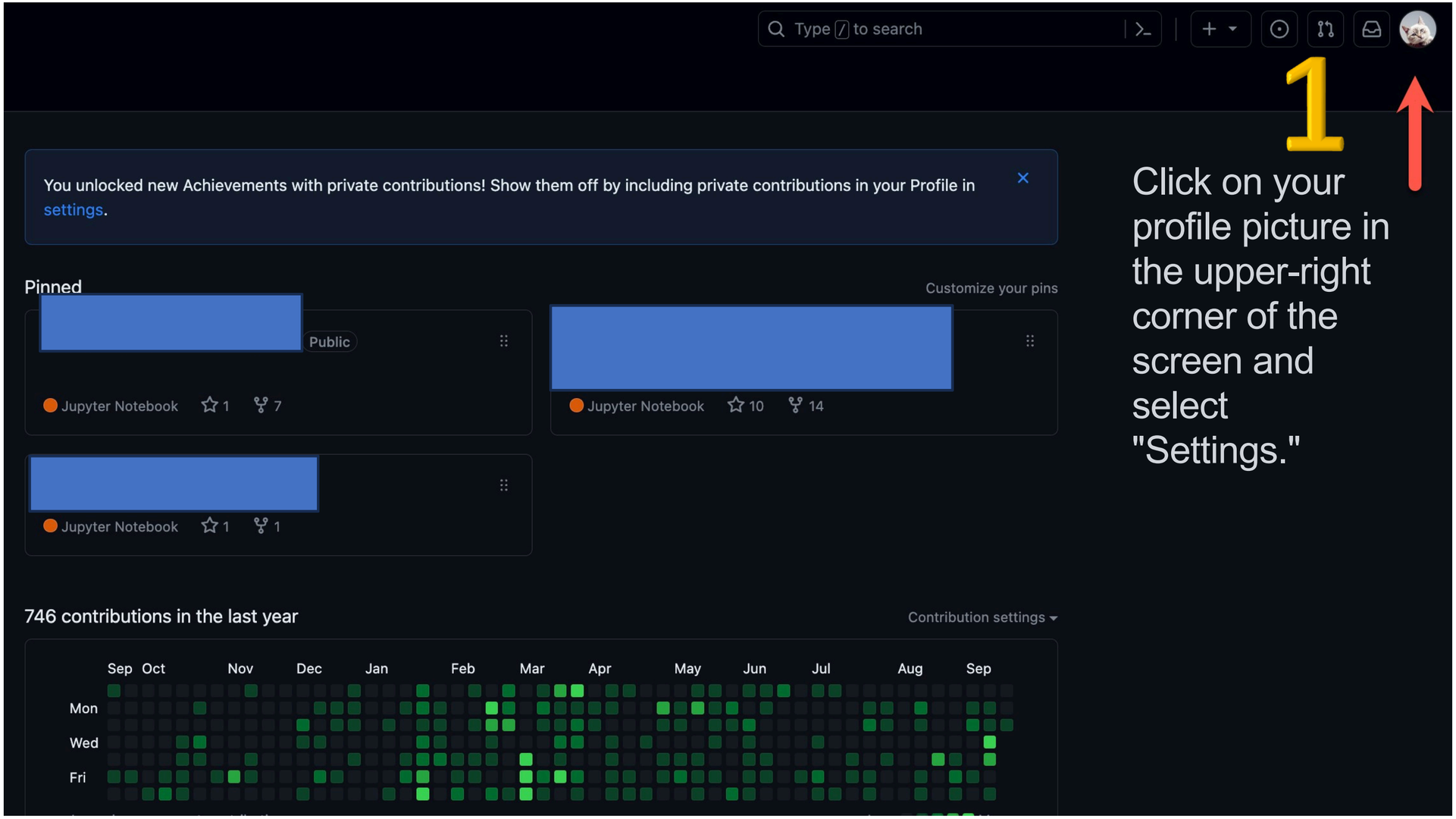
- Clone Syntax with PAT:

git clone https://username:YOUR_TOKEN@github.com/username/repo.git

- Previously we used:

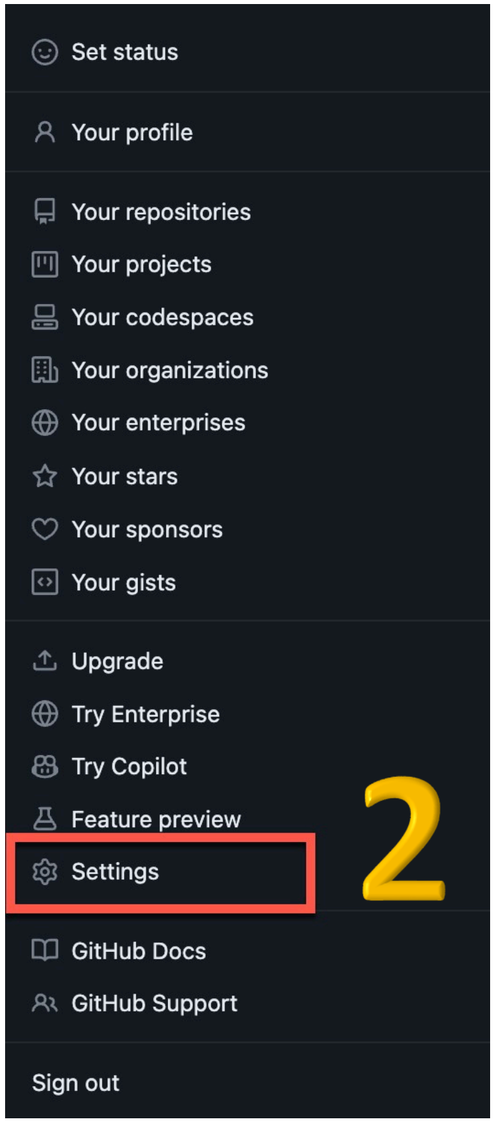
git clone https://github.com/account/repo.git

Create a Personal Access Token



1

Click on your profile picture in the upper-right corner of the screen and select "Settings."



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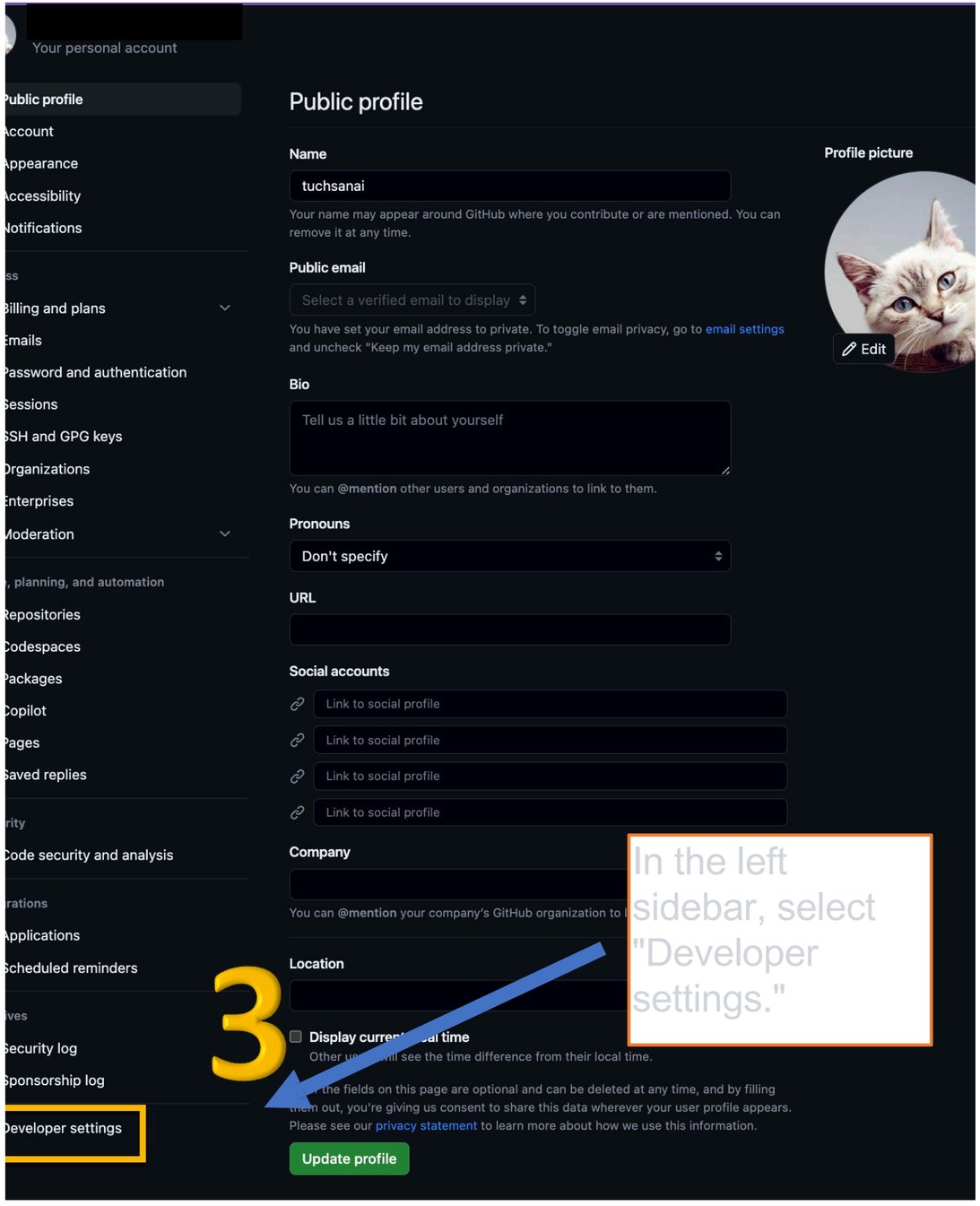
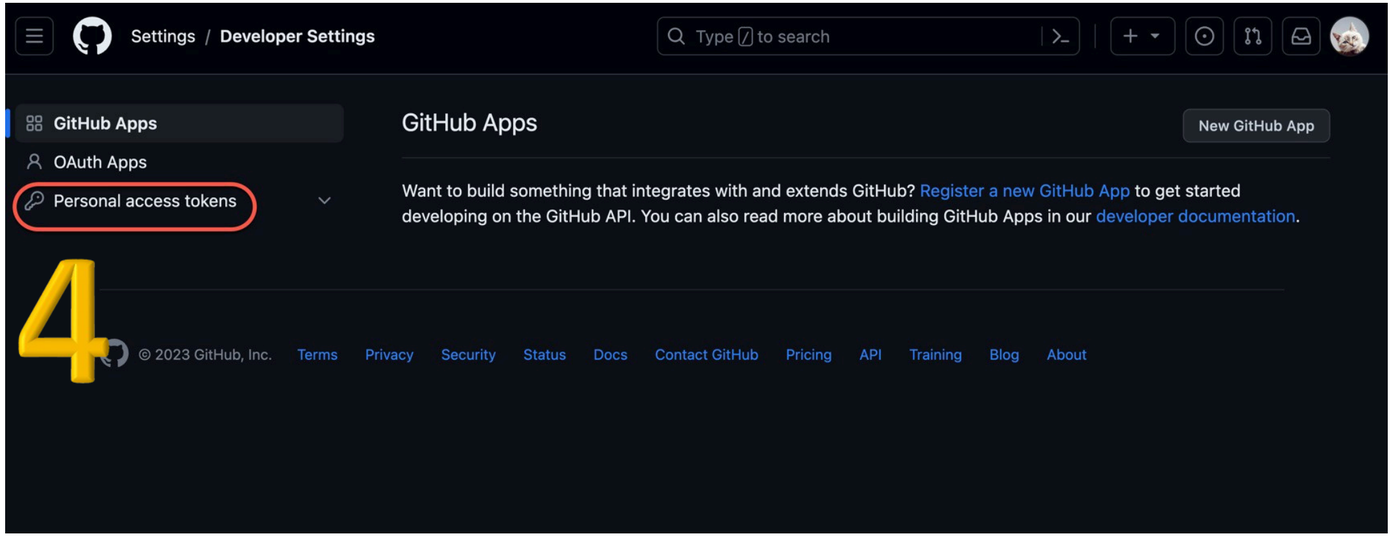

Your personal account
Public profile
Account
Appearance
Accessibility
Notifications
Billing and plans
Emails
Password and authentication
Sessions
SSH and GPG keys
Organizations
Enterprises
Moderation
Repositories
Codespaces
Packages
Copilot
Pages
Saved replies
Code security and analysis
Applications
Scheduled reminders
Security log
Sponsorship log
Developer settings
Public profile
Name
tuchsanai
Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.
Public email
Select a verified email to display
You have set your email address to private. To toggle email privacy, go to email settings and uncheck "Keep my email address private."
Bio
Tell us a little bit about yourself
You can @mention other users and organizations to link to them.
Pronouns
Don't specify
URL
Social accounts
Link to social profile
Company
Location
Display current local time
Other users will see the time difference from their local time.
All the fields on this page are optional and can be deleted at any time, and by filling them out, you're giving us consent to share this data wherever your user profile appears. Please see our privacy statement to learn more about how we use this information.
Update profile
Profile picture
Edit
3
In the left sidebar, select "Developer settings."


Settings / Developer Settings
Type / to search
GitHub Apps
OAuth Apps
Personal access tokens
GitHub Apps
New GitHub App
Want to build something that integrates with and extends GitHub? Register a new GitHub App to get started developing on the GitHub API. You can also read more about building GitHub Apps in our developer documentation.
4
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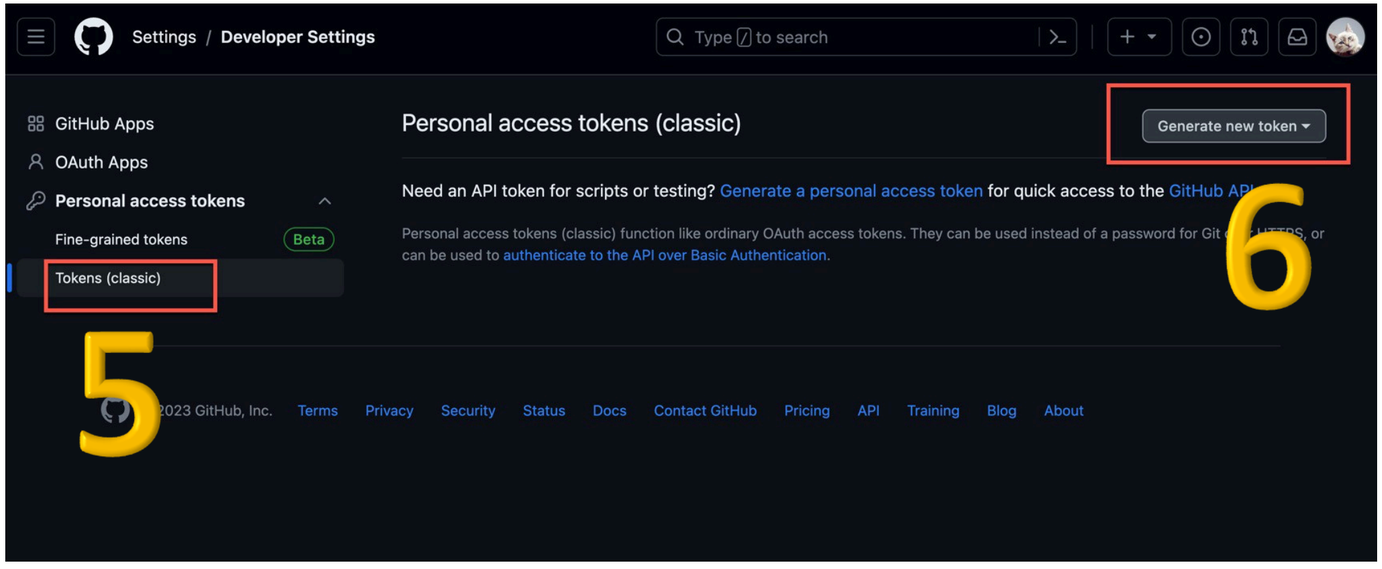

Settings / Developer Settings
Type / to search
GitHub Apps
OAuth Apps
Personal access tokens
Fine-grained tokens
Beta
Tokens (classic)
Personal access tokens (classic)
Generate new token
Need an API token for scripts or testing? Generate a personal access token for quick access to the GitHub API.
Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.
5
6
2023 GitHub, Inc. Terms Privacy Security Status Docs Contact GitHub Pricing API Training Blog About

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.

Note

test

7

What's this token for?

Expiration *

7 days

The tok… expire on Tue, Sep 26 2023

8

Select scopes

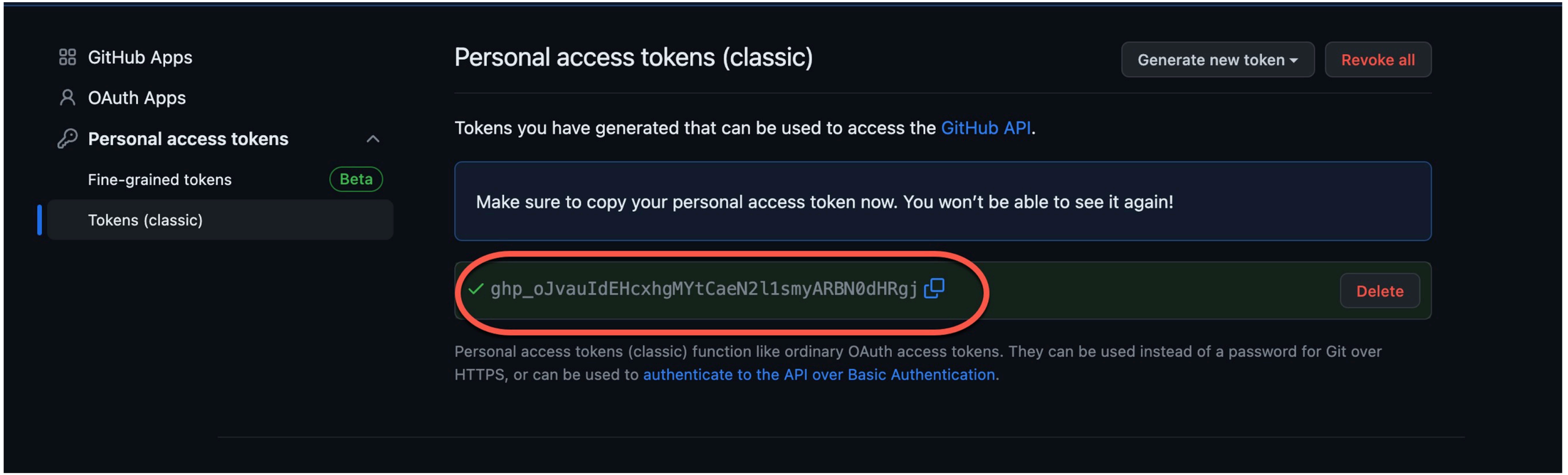
Scopes define the access for personal tokens. Read more about OAuth scopes.

repo	Full control of private repositories
repo:status	Access commit status
repo_deployment	Access deployment status
public_repo	Access public repositories
repo:invite	Access repository invitations
security_events	Read and write security events

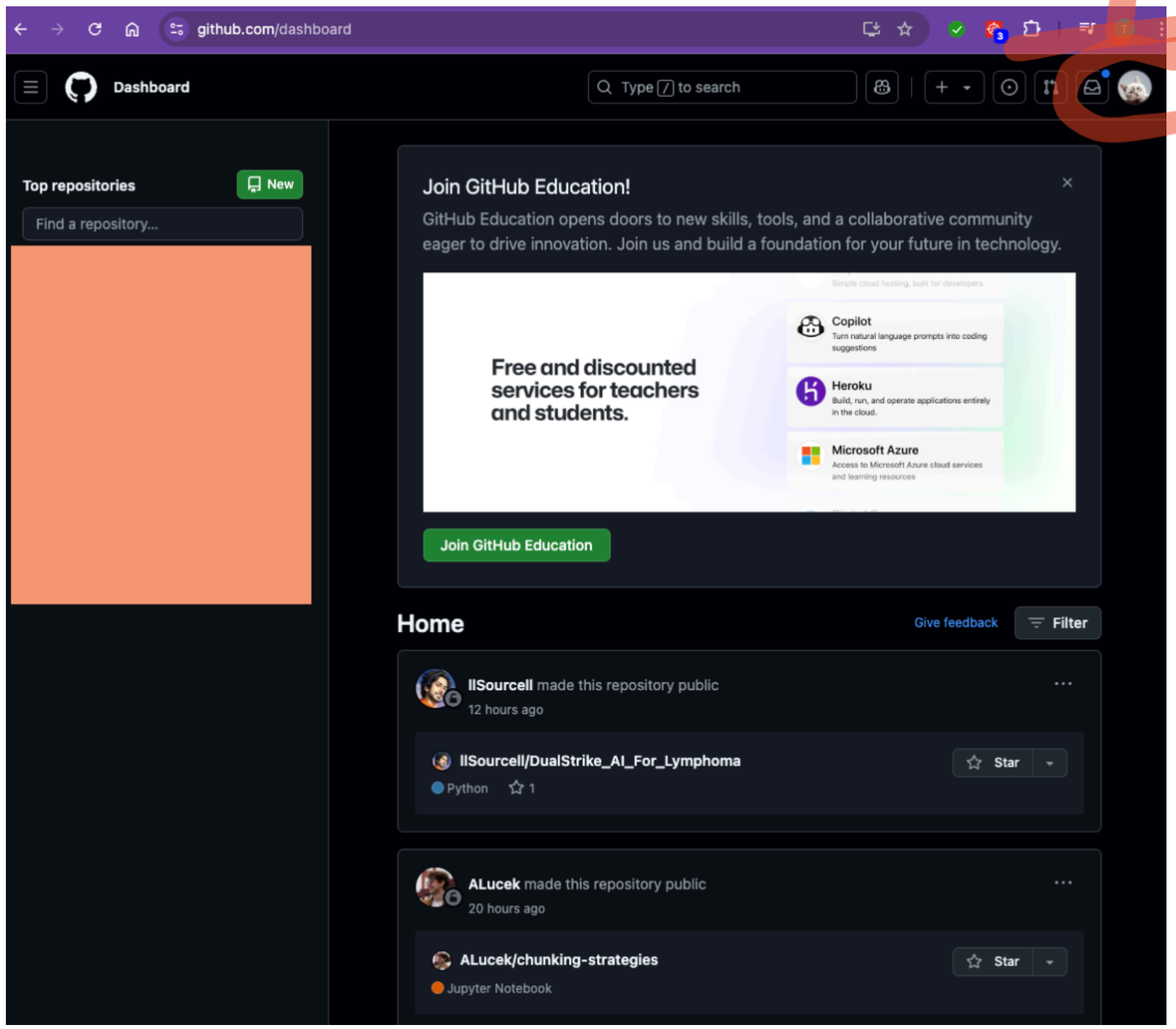
9

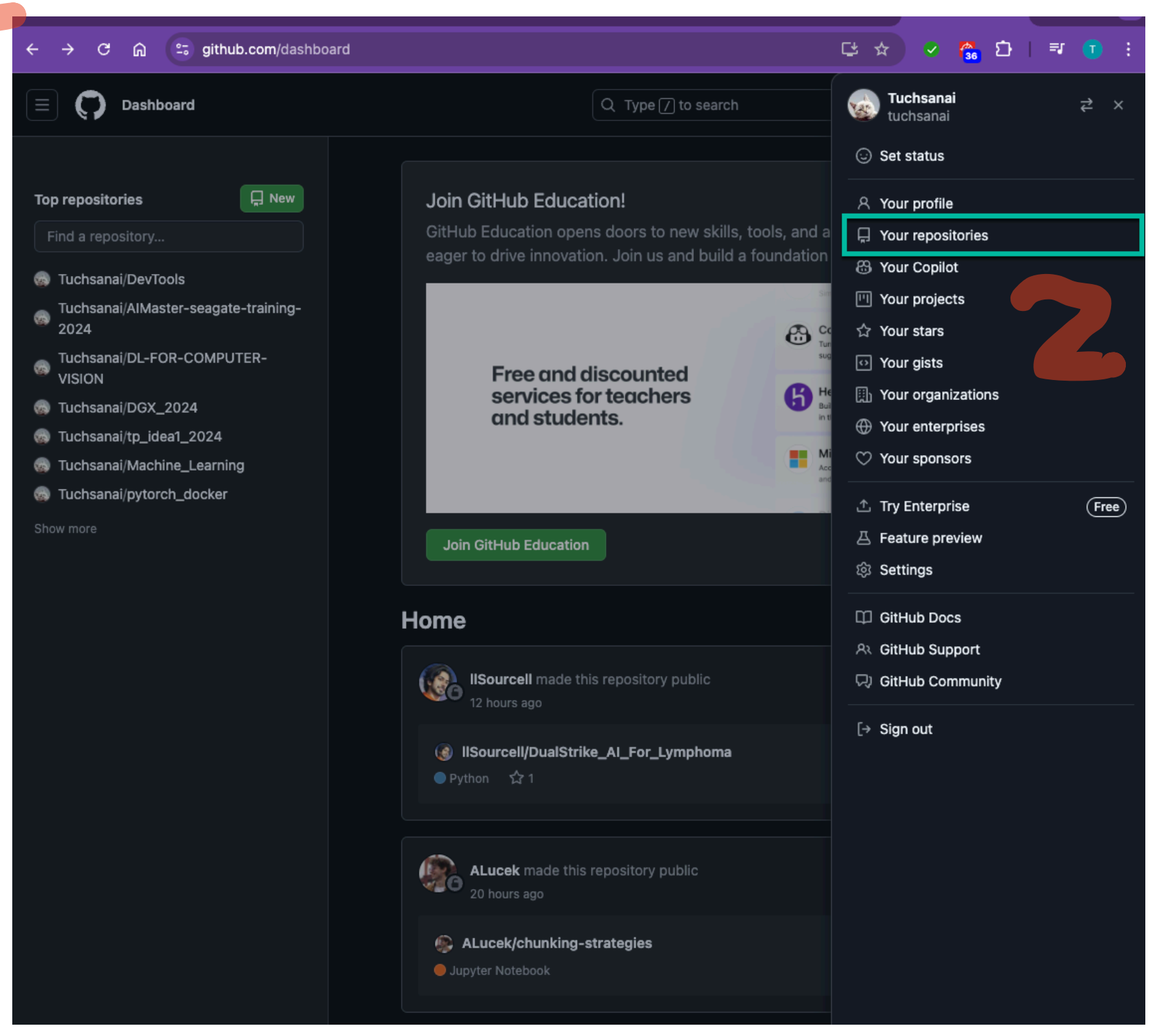
repo	Full control of private repositories
repo:status	Access commit status
repo_deployment	Access deployment status
public_repo	Access public repositories
repo:invite	Access repository invitations
security_events	Read and write security events
workflow	Update GitHub Action workflows
write:packages	Upload packages to GitHub Package Registry
read:packages	Download packages from GitHub Package Registry
delete:packages	Delete packages from GitHub Package Registry
admin:org	Full control of orgs and teams, read and write org projects
write:org	Read and write org and team membership, read and write org projects
read:org	Read org and team membership, read org projects
manage_runners:org	Manage org runners and runner groups
admin:public_key	Ful… ontrol of user public keys
write:public_key	Wr… user public keys
read:public_key	R… user public keys
admin:repo_hook	Full control of repository hooks
write:repo_hook	Write repository hooks
read:repo_hook	Read repository hooks
admin:org_hook	Full control of organization hooks
gist	Create gists
notifications	Access notifications
user	Update ALL user data
read:user	Read ALL user profile data
user:email	Access user email addresses (read-only)
user:follow	Follow and unfollow users

10

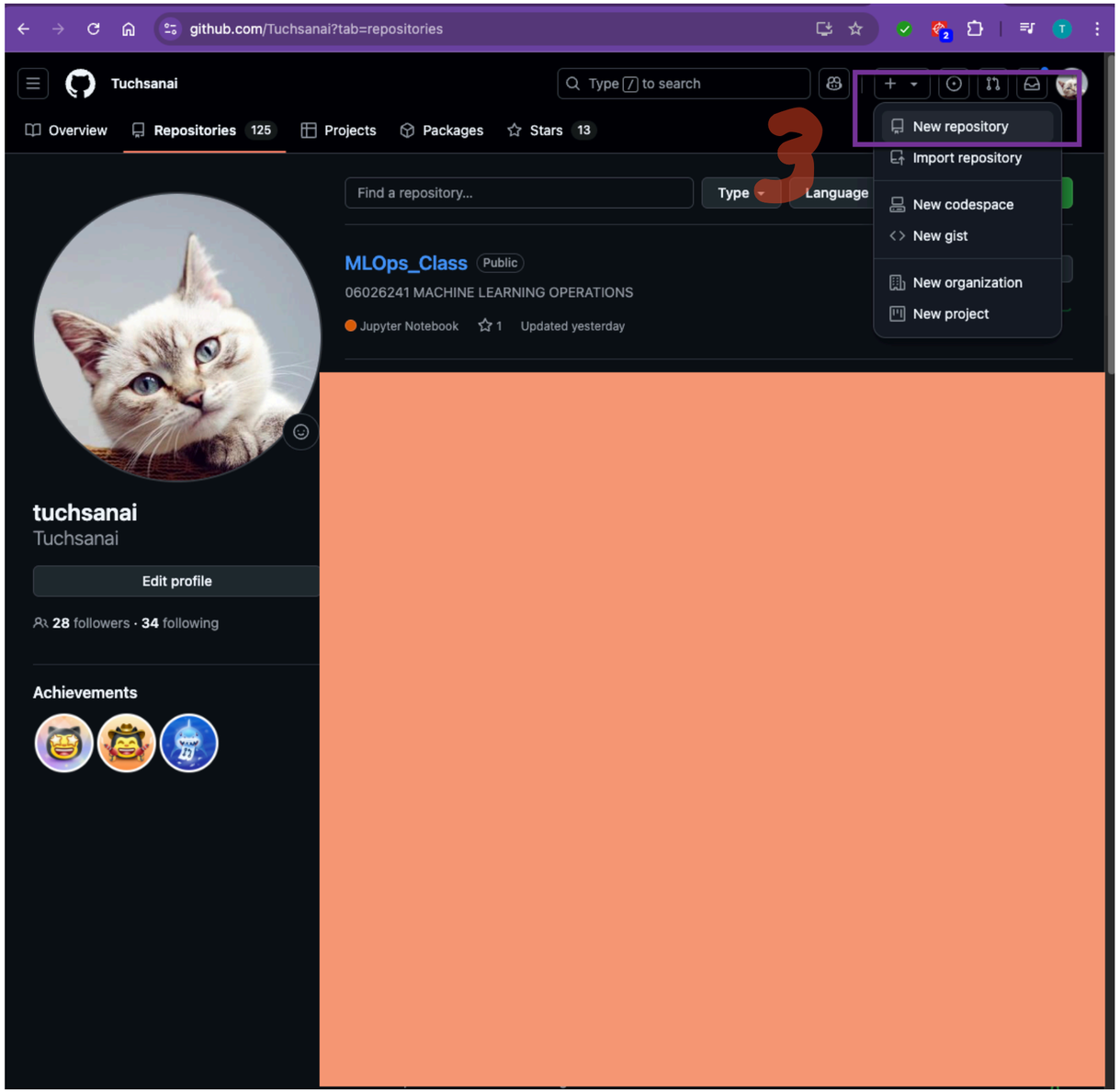

GitHub Apps
OAuth Apps
Personal access tokens
Fine-grained tokens
Beta
Tokens (classic)
Personal access tokens (classic)
Generate new token
Revoke all
Tokens you have generated that can be used to access the GitHub API.
Make sure to copy your personal access token now. You won't be able to see it again!
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
Delete
Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.

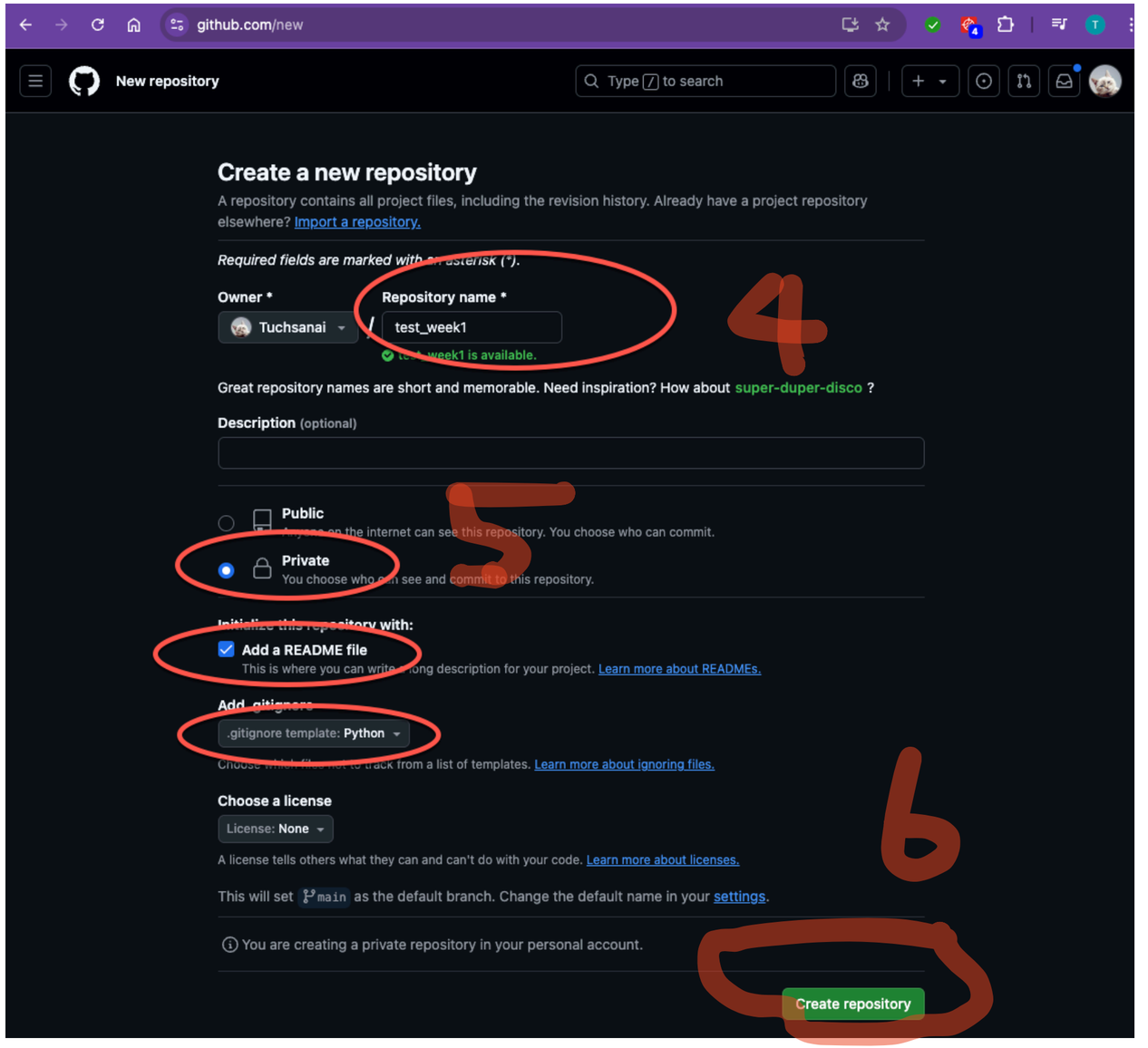
Create a new Private Repository


github.com/dashboard
Dashboard
Type / to search
Top repositories
New
Find a repository...
Join GitHub Education!
GitHub Education opens doors to new skills, tools, and a collaborative community eager to drive innovation. Join us and build a foundation for your future in technology.
Free and discounted services for teachers and students.
Copilot
Turn natural language prompts into coding suggestions
Heroku
Build, run, and operate applications entirely in the cloud.
Microsoft Azure
Access to Microsoft Azure cloud services and learning resources
Join GitHub Education
Home
Give feedback
Filter
llSourcell made this repository public
12 hours ago
llSourcell/DualStrike_AI_For_Lymphoma
Python
1
Star
ALucek made this repository public
20 hours ago
ALucek/chunking-strategies
Jupyter Notebook
Star

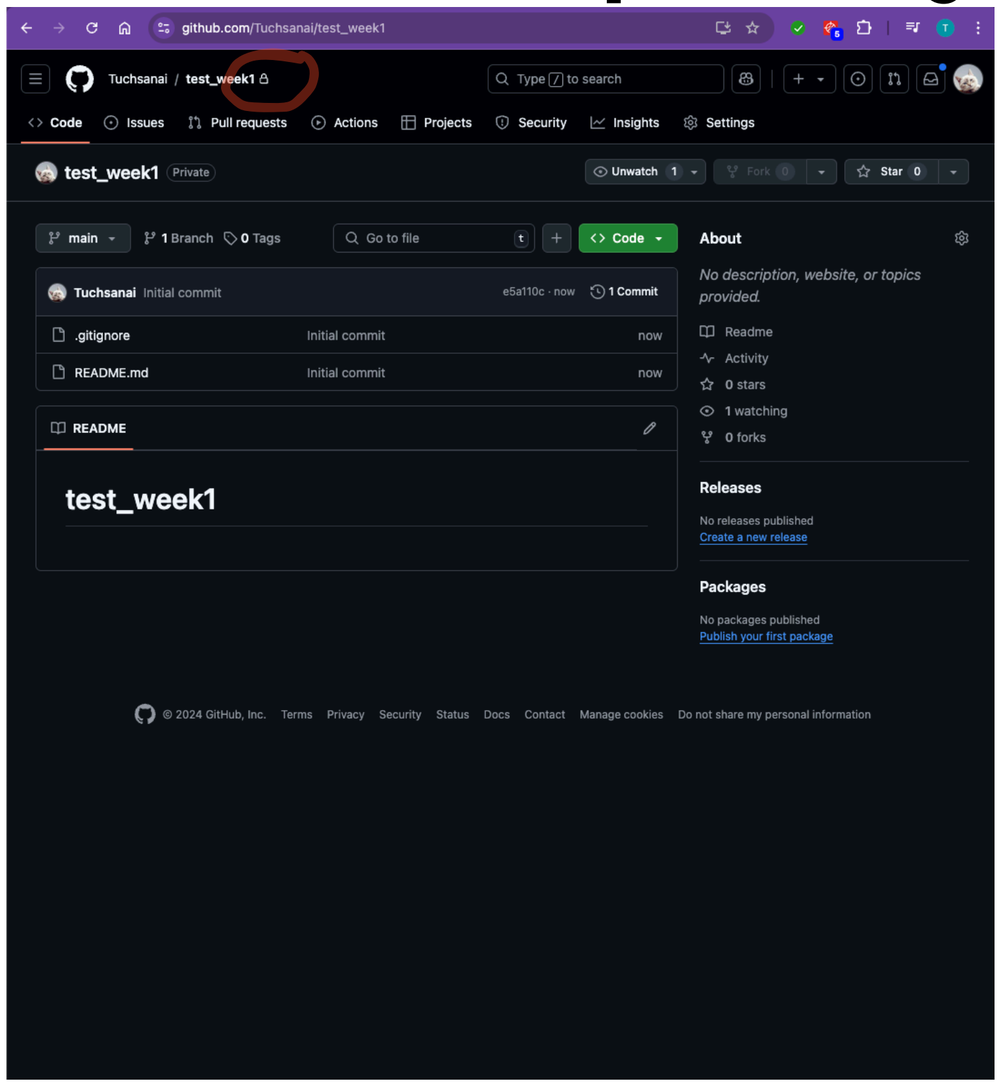

github.com/dashboard
Dashboard
Type / to search
Top repositories
New
Find a repository...
Tuchsanai/DevTools
Tuchsanai/AIMaster-seagate-training-2024
Tuchsanai/DL-FOR-COMPUTER-VISION
Tuchsanai/DGX_2024
Tuchsanai/tp_idea1_2024
Tuchsanai/Machine_Learning
Tuchsanai/pytorch_docker
Show more
Join GitHub Education!
Free and discounted services for teachers and students.
Join GitHub Education
Home
llSourcell made this repository public
12 hours ago
llSourcell/DualStrike_AI_For_Lymphoma
Python
1
ALucek made this repository public
20 hours ago
ALucek/chunking-strategies
Jupyter Notebook
Tuchsanai
tuchsanai
Set status
Your profile
Your repositories
Your Copilot
Your projects
Your stars
Your gists
Your organizations
Your enterprises
Your sponsors
Try Enterprise
Free
Feature preview
Settings
GitHub Docs
GitHub Support
GitHub Community
Sign out

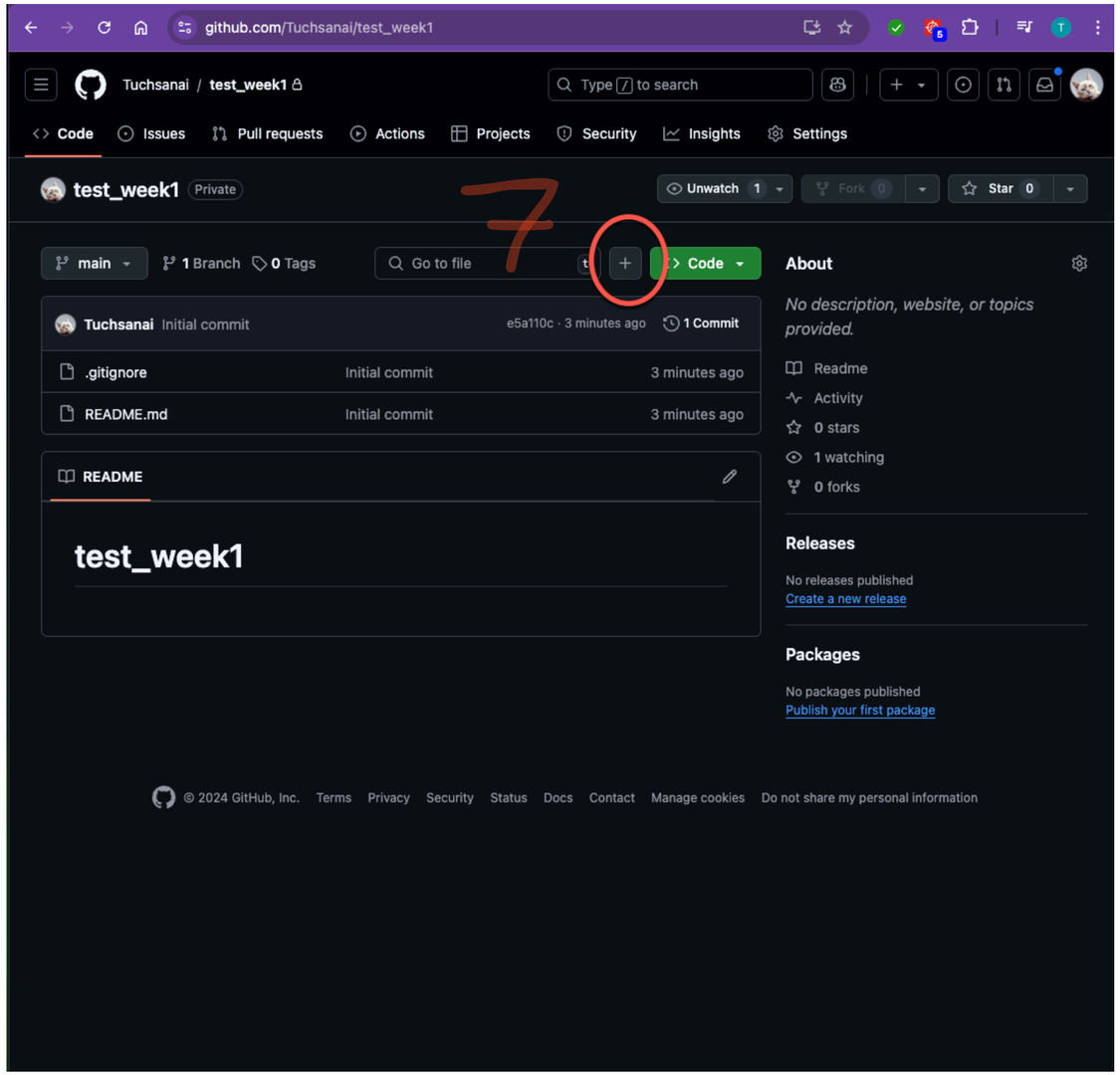
Click New repository



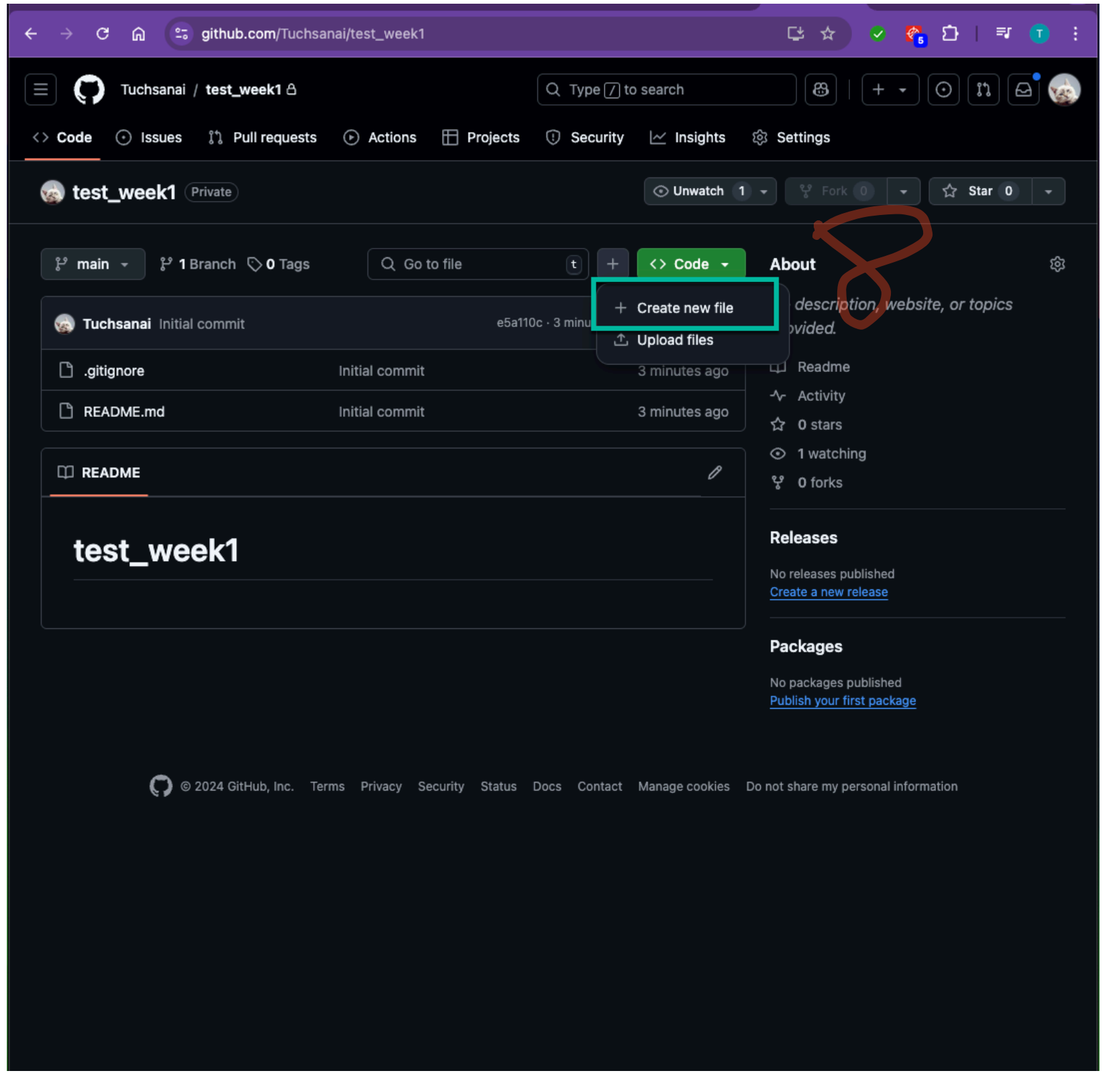


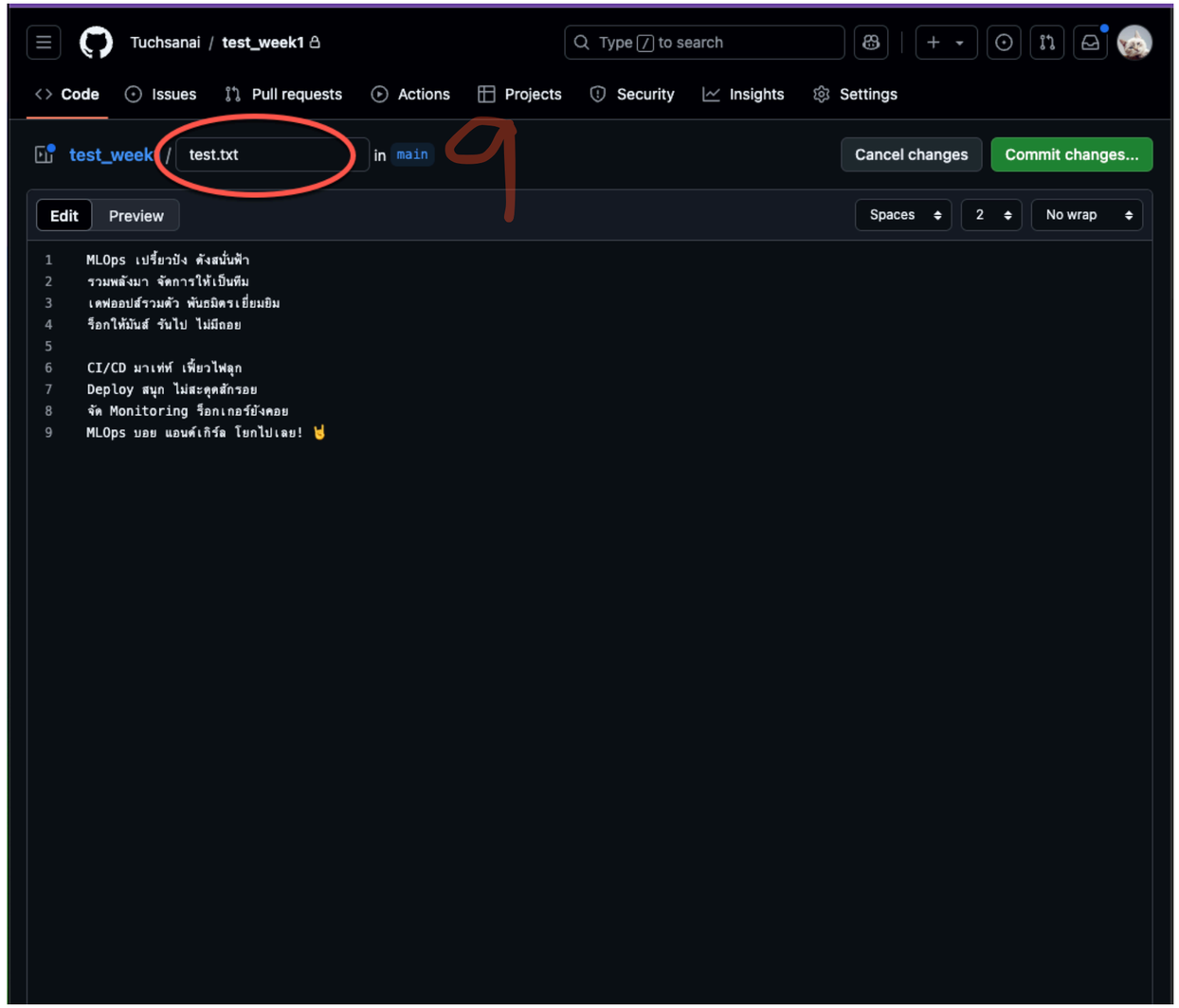
Privated repository

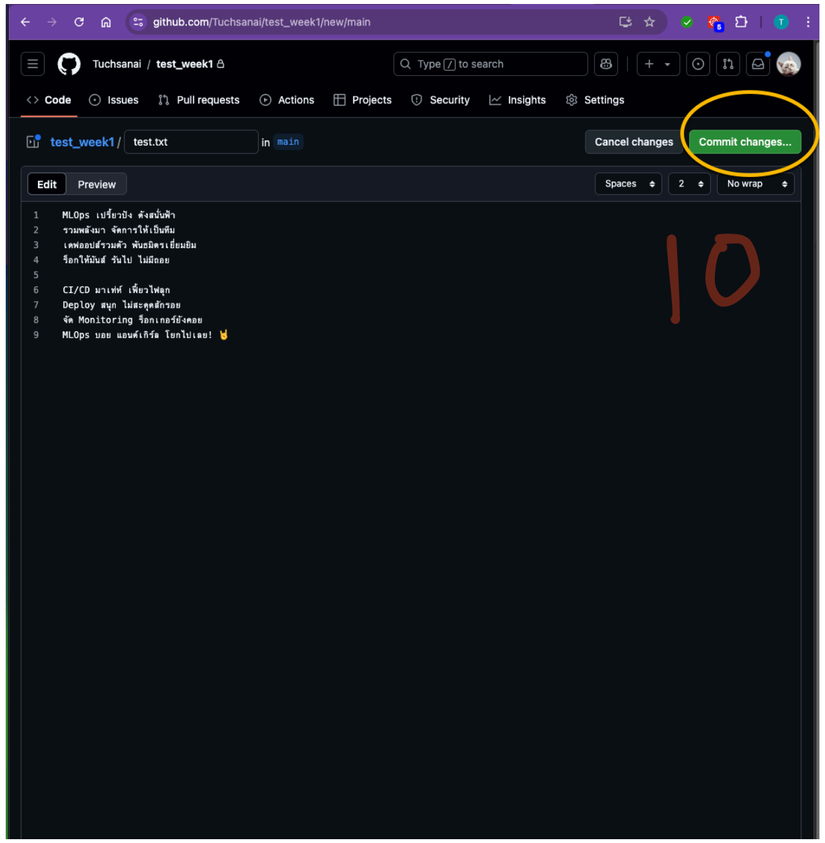


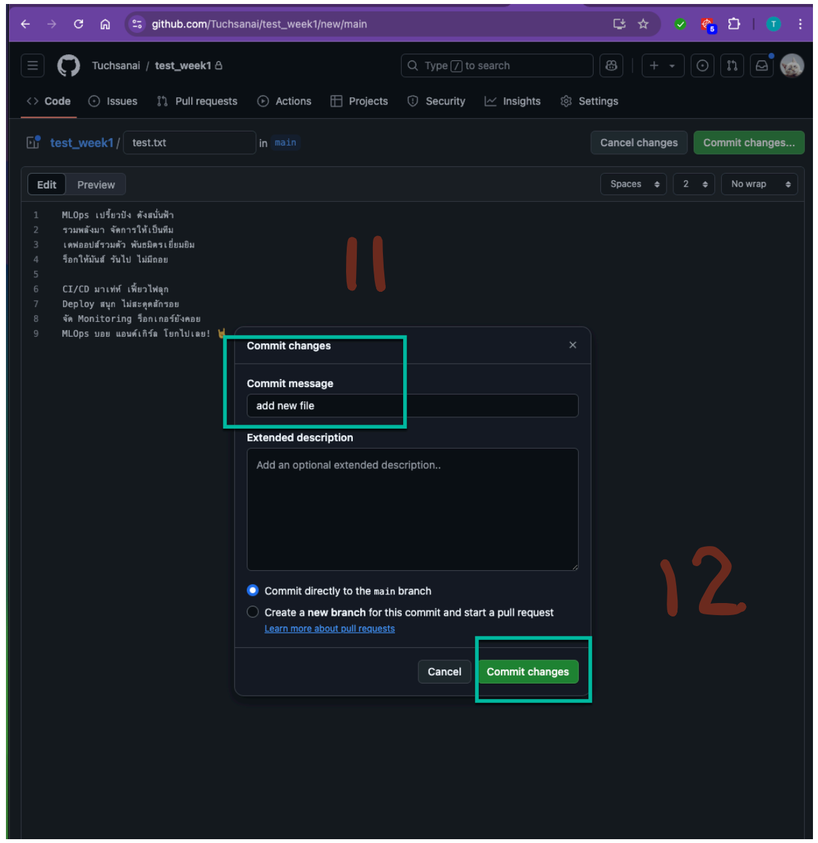


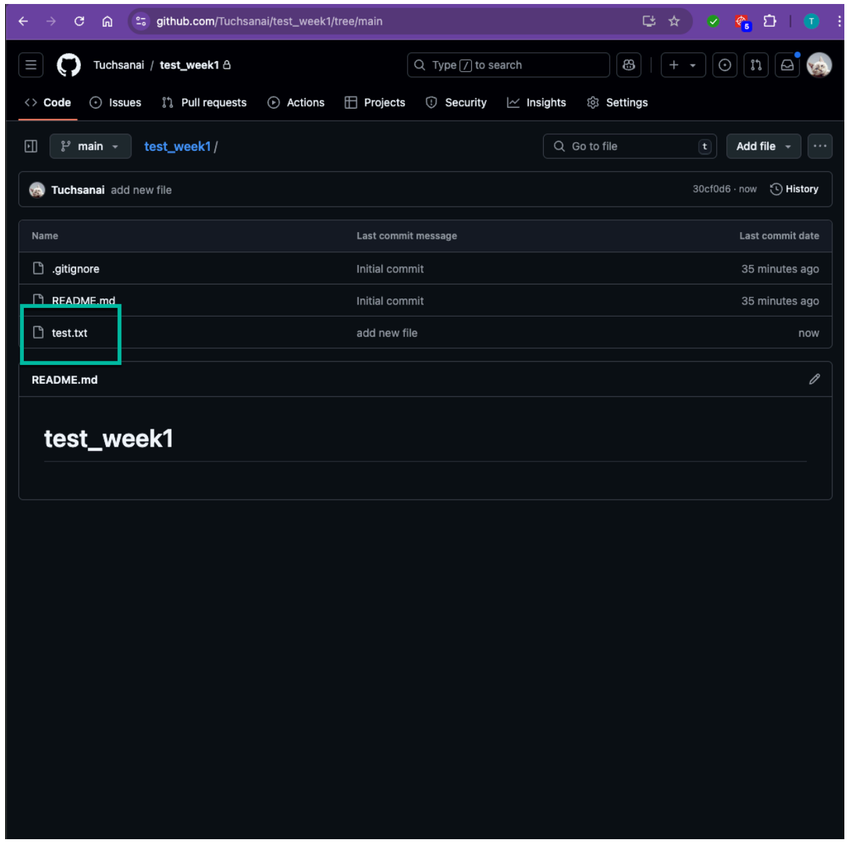
สร้าง file : test.txt











- Clone private respository:

git clone https://username:YOUR_TOKEN@github.com/username/repo.git

สรุปคำสั่งสำคัญ (Cheat Sheet)

```
# ── ตรวจสอบ / ตั้งค่า ──────────────────
git --version                       # เช็กว่ามี Git และเวอร์ชันอะไร
git config --global user.name "name"
git config --global user.email "email"
git config --global credential.username "user"
git config --list                       # ดู config ทั้งหมด (= git config -l)
git config --list --show-origin         # ดูว่าค่ามาจากไฟล์ไหน

# ── ลบ config / origin เก่า ──────────
git config --global --unset user.name
git config --global --unset user.email
git remote remove origin

# ── สร้าง / ตรวจสอบ repo ──────────────
git init                            # สร้าง repo (เกิดโฟลเดอร์ .git)
git status                          # ดูสถานะปัจจุบัน

# ── clone ─────────────────────────────
git clone https://github.com/user/repo.git
git clone https://user:TOKEN@github.com/user/repo.git  # private
```